Numerical Methods for Elliptic Problems

Lecture Notes for the Winter Semester 2026/27

Michael Neunteufel

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Johannes Kepler University Linz

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Preface

These lecture notes are based on Herbert Egger's notes from the summer semester 2022. The material has been revised and reorganized for the present course.

Corrections, modifications, and additional material in this version are the responsibility of the adapting author.

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Introduction

This course is about elliptic variational problems, stemming from boundary value problems for certain partial differential equations, and their systematic numerical approximation by the finite element method. Most of the theory will be developed for a simple system, i.e., the Dirichlet problem for the Poisson equation

$$\begin{aligned} -\operatorname{div}(a\nabla u) &= f \qquad \text{in } \Omega, \\ u &= 0 \qquad \text{on } \partial\Omega. \end{aligned}$$

The results and methods, however, carry over quite naturally to much more general problems, e.g., in heat and mass transfer, mechanics, fluid flow and electromagnetics, making the finite element method a very powerful tool for the simulation of complex physical processes in science and engineering.

During the lecture, we will touch on various aspects that are important for the development and analysis of discretization methods:

- Modelling: study the physical meaning of the equations and the mathematical tools underlying their derivation.
- Analysis: establish existence, uniqueness and stability of solutions.
- Discretization: develop tools for the systematic numerical approximation.
- Numerical analysis: analyse discretized problems and prove convergence.
- Implementation: study the basic approach to an efficient realization and derive complexity estimates for the numerical solution.

Literature. For preparation of this lecture, we used material from standard textbooks on finite element methods, in particular

- D. Braess: Finite Elements: Theory, Fast Solvers, and Applications in Solid Mechanics. Cambridge University Press, 2007.

as well as

- P. Ciarlet: The Finite Element Method for Elliptic Problems. SIAM, 2002.
- S. Brenner and L. R. Scott: The Mathematical Theory of Finite Element Methods. Springer 2008.
- A. Ern and J.-L. Guermond: Theory and Practice of Finite Elements. Springer, 2004.
- S. Bartels: Numerical Approximation of Partial Differential Equations. Springer 2016.

Further references for particular results will be given during the course.

1 Model problems and principles

Before we delve into the details of the mathematical analysis and numerical approximation of elliptic boundary value problems for partial differential equations, we want to highlight some physical principles and mathematical arguments used for their derivation and introduce some typical model problems that lie in the wider scope of the lecture.

1.1 Mass and heat transfer

We consider the transport of a chemical substance, a tracer, in the flow of a fluid, e.g., water or air. By the principle of *conservation of matter*, we obtain the balance equation (change of total amount = flow over boundary + sources/sinks)

$$\frac{d}{dt}\int_\omega u(x,t)\,dx = -\int_{\partial\omega} \vec{q}(x,t)\cdot\vec{n}(x)\,ds(x) + \int_\omega s(x,t)\,dx,$$

which holds for every control volume ω under consideration. Here u is the concentration of the substance, $\vec{q}$ the flux, $\vec{n}$ the outer normal on $\partial\omega$, and s a source term. Under steady state conditions, all quantities can be assumed independent of time. Using the Gauß-integral law

$$\int_\omega \operatorname{div}(\vec{q})\,dx = \int_{\partial\omega} \vec{q}\cdot\vec{n}\,ds,$$

and dropping the left hand side in the first equation, we arrive at

$$\int_\omega \operatorname{div}(\vec{q})\,dx = \int_\omega s\,dx,$$

which still holds for all control volumes ω under consideration. This can be turned into a partial differential equation by the following result.

Lemma 1.1 (Fundamental lemma 1)**.**
Let $\Omega \subset \mathbb{R}^d$ be some open set and $f \in C(\overline{\Omega})$. Then

$$\int_\omega f(x)\,dx = 0 \quad \forall \omega \subset \Omega \qquad \Leftrightarrow \qquad f(x) = 0 \quad \forall x \in \Omega.$$

It suffices to restrict the control volume ω to closed balls lying in Ω.

The proof is left as an exercise. Note that a similar statement holds for integrable $f \in L^1(\Omega)$. Then the right identity holds for almost every (a.e.) $x \in \Omega$.

By use of the fundamental lemma, we directly obtain the following balance equation in differential form

$$\operatorname{div}(\vec{q}) = s \qquad \text{in } \Omega,$$

where $\Omega \subset \mathbb{R}^d$ is the *computational domain* which we consider. In order to recover information about the concentration u, we have to prescribe how the flux $\vec{q}$ and the source density s depend on u. For this we use the *constitutive equations*

$$\vec{q} = -a\nabla u + \vec{b}u \qquad \text{and} \qquad s = f - cu,$$

where a is a diffusion coefficient, $\vec{b}$ the flow velocity of the fluid, f an external source, and c a reaction coefficient. The first term in the flux amounts to diffusion (=Fick's law) and the second to transport (=Transport theorem). The source term says that substance is either injected/subtracted directly or degrades through some chemical process.

By inserting the constitutive laws into the balance equation, we arrive at the *diffusion-convection-reaction* equation

$$-\operatorname{div}(a\nabla u - \vec{b}u) + cu = f \qquad \text{in } \Omega. \tag{1.1}$$

In order to fully describe the concentration profile in steady state, we additionally require *boundary conditions*, e.g.

$$u = g \qquad \text{on } \Gamma_D \tag{1.2}$$

$$\vec{n} \cdot (a\nabla u - \vec{b}u) = h \qquad \text{on } \Gamma_N = \partial\Omega \setminus \Gamma_D. \tag{1.3}$$

The first (=Dirichlet) condition directly prescribes the concentration, while the second (=Neumann) condition fixes the normal flux at the boundary. Note that we prescribe exactly one condition at every point of the boundary.

Model problem

The system (1.1)–(1.3) makes up a *boundary value problem* for a *linear elliptic partial differential equation of second order* (short: an elliptic bvp) which describes the distribution of a chemical substance in the steady flow of a fluid. By the very same arguments, we could also consider the transfer of heat and arrive at the stationary heat equation. If we assume $\Gamma_N = \emptyset$, $\vec{b} = 0$, $c = 0$, and $g = 0$, the above system further simplifies to

$$-\operatorname{div}(a\nabla u) = f \qquad \text{in } \Omega, \tag{1.4}$$

$$u = 0 \qquad \text{on } \partial\Omega, \tag{1.5}$$

which will serve as the basic model problem for most of our considerations. Another interesting case occurs, if we neglect diffusion and set $a = 0$. Then we obtain a pure transport problem, which is a linear hyperbolic differential equation of first order. We will come back to this problem at a later stage.

1.2 Minimal surface problem

We consider a thin membrane which is stretched over a fixed frame, e.g., a wire loop. The position of the frame is given by a height function $g : \partial\Omega \to \mathbb{R}$, where $\Omega \subset \mathbb{R}^2$ is the projection of the area enclosed by the frame onto a flat surface. Now let $u : \Omega \to \mathbb{R}$ denote the vertical position of the membrane. We assume that in equilibrium, the membrane takes the form of minimal surface area, i.e.,

$$A(u) := \int_\Omega \sqrt{1 + |\nabla u(x)|^2}\, dx \to \min_{u : u|_{\partial\Omega} = g}.$$

Some restrictions on the possible deformations (=deflections) described by u are, of course, required to ensure that the integral is well-defined and, eventually, to guarantee the existence of a minimizer. If u is a sufficiently regular minimizer, then certainly $A(u+tv) \geq A(u)$ for all $t \in \mathbb{R}$ and all admissible variations v which satisfy $v|_{\partial\Omega} = 0$. The latter condition is required to ensure that $u + tv$ again is tied to the frame at the boundary. Let us denote by $a(w) = \sqrt{1 + |w|^2}$, i.e., w stands for ∇u in the formula for the surface area. Then by Taylor expansion, we see that $a(w + \delta w) = a(w) + a'(w) \cdot \delta w + O(|\delta w|^2)$ with derivative (=Jacobian) $a'(w) = \frac{w}{\sqrt{1+|w|^2}}$. With this, we can formally compute

$$\begin{aligned}
\delta A(u)[v] :&= \lim_{t\to 0} \frac{1}{t}(A(u + tv) - A(u)) \\
&= \lim_{t\to 0} \frac{1}{t} \int_\Omega \big(a(\nabla u + t\nabla v) - a(\nabla u)\big)\, dx = \int_\Omega a'(\nabla u) \cdot \nabla v\, dx \\
&= \int_\Omega \frac{1}{\sqrt{1 + |\nabla u|^2}} \nabla u \cdot \nabla v\, dx \qquad \forall v : v|_{\partial\Omega} = 0.
\end{aligned}$$

This is called the *first variation* of the functional $A(u)$ in direction v, and corresponds to a directional derivative of the map $A : u \mapsto A(u)$. By our previous considerations, we see that $\delta A(u)[v] = 0$ for all admissible v, and hence

$$\int_\Omega \frac{1}{\sqrt{1 + |\nabla u|^2}} \nabla u \cdot \nabla v = 0 \qquad \forall v : v|_{\partial\Omega} = 0.$$

Since the test function v denotes a variation of u, this identity is called a *variational equation*, and likewise the corresponding minimization problem is called a

variational problem. Such infinite-dimensional minimization problems are studied in the context of the *calculus of variations.*

Let us proceed by recalling the integration-by-parts formula

$$\int_\Omega \vec{q} \cdot \nabla v \, dx = -\int_\Omega \operatorname{div}(\vec{q})\, v \, dx + \int_{\partial\Omega} \vec{n} \cdot \vec{q}\, v \, ds,$$

which follows from the Gauß-integral law and the chain rule; see the exercises. This allows us to rewrite the variational equation into the equivalent form

$$-\int_\Omega \operatorname{div}\left(\frac{1}{\sqrt{1+|\nabla u|^2}}\nabla u\right) v \, dx = 0 \qquad \forall v : v|_{\partial\Omega} = 0.$$

Note that the boundary term in the integration-by-parts formula vanishes, since we have assumed $v|_{\partial\Omega} = 0$. To proceed further, we use the following observation.

Lemma 1.2 (Fundamental lemma 2)**.**
Let $\Omega \subset \mathbb{R}^d$ be open and $f \in C(\overline{\Omega})$. Then

$$\int_\Omega f\, v \, dx = 0 \quad \forall v \in C(\overline{\Omega}) : v|_{\partial\Omega} = 0 \qquad \Leftrightarrow \qquad f(x) = 0 \quad \forall x \in \Omega.$$

The proof is again left as an exercise. We note that the statement can again be generalized concerning the regularity of the functions f and v.

Applying this result to the above equation immediately leads to the nonlinear elliptic partial differential equation

$$-\operatorname{div}\left(\frac{1}{\sqrt{1+|\nabla u|^2}}\nabla u\right) = 0 \qquad \text{in } \Omega. \tag{1.6}$$

From the fact that the membrane is attached to the frame, we obtain the additional boundary condition

$$u = g \qquad \text{on } \partial\Omega. \tag{1.7}$$

The system (1.6)–(1.7) describes the vertical position u of a membrane which is stretched over a rigid frame and which has the minimal surface area.

Linearization and the membrane problem

If $g \approx 0$, we have $u \approx 0$ and we may further assume the small-slope regime $|\nabla u| \ll 1$. As a consequence, we see that $\sqrt{1+|\nabla u|^2} \approx 1$, and we obtain the corresponding linearized problem

$$\begin{aligned} -\Delta u &= 0 \qquad \text{in } \Omega \\ u &= g \qquad \text{on } \partial\Omega, \end{aligned}$$

which is called the *membrane equation* plus boundary conditions. Reverting the arguments used in the derivation of (1.6)–(1.7), one can show that solutions of this problem are minimizers of the quadratic energy

$$E(u) = \frac{1}{2}\int_\Omega |\nabla u|^2 dx \to \min_{u:u|_{\partial\Omega}=g}.$$

Details are again left as an exercise. If we choose a right hand side f in the membrane equation and set $g = 0$ at the boundary, we arrive at our model problem (1.4)–(1.5) with $a = 1$. This system then models the deflection of a horizontal membrane under the action of a vertical distributed load f.

1.3 The biharmonic problem

The mechanical forces within a membrane are due to stretching and they act tangentially to the surface of the membrane. If we instead consider a thin plate, then the main forces are due to bending of the plate. Neglecting other effects and setting the material parameters to one, the deflection of a horizontal plate under the action of a vertical force f is described by the biharmonic equation

$$\operatorname{div}\operatorname{div}(\nabla^2 u) = \Delta^2 u = f \qquad \text{in } \Omega, \tag{1.8}$$

which is a linear partial differential equation of fourth order. This equation describes the equilibrium of internal and external forces in the *Kirchhoff-Love plate theory*; see Lecture: Continuum Mechanics. To fully describe the position of the plate, we additionally require appropriate boundary conditions, e.g.

$$u = 0 \qquad \text{and} \qquad \partial_n u = 0 \qquad \text{on } \partial\Omega, \tag{1.9}$$

which describe that the plate is clamped. Note that we now have two conditions at every point of the boundary. Since by the first condition, also the tangential derivative along the boundary vanishes, we conclude that even $\nabla u = 0$ on $\partial\Omega$.

With very similar arguments as used in the derivation of the minimal surface and membrane equation, one can show that the equilibrium position u described by the system (1.8)–(1.9) corresponds to a minimum of, see exercises,

$$\frac{1}{2}\int_\Omega |\Delta u|^2\, dx - \int_\Omega f\, u\, dx \to \min_{u:u|_{\partial\Omega}=\partial_n u|_{\partial\Omega}=0}.$$

The equilibrium of forces is thus again tightly related to the minimization of a certain energy. Apart from the appearance of Δ instead of ∇ and the two boundary conditions instead of one, this problem has a very similar form as the membrane equation and as our model problem (1.4)–(1.5) and, in fact, can be treated with very similar arguments, both, analytically and numerically.

1.4 Linear elasticity

We now turn to the deformation of more general elastic bodies. In that case, the equilibrium of forces is expressed as

$$-\operatorname{div}(\sigma) = \vec{f} \qquad \text{in } \Omega.$$

In the case of small deformations, we may identify $\Omega \subset \mathbb{R}^3$ with the reference configuration of the object under consideration. Furthermore, $\vec{f} : \Omega \to \mathbb{R}^3$ is the density of external forces, and $\sigma : \Omega \to \mathbb{R}^{3\times 3}$ is a symmetric tensor field, the so-called stress tensor. The divergence operator acts in a row by row fashion, i.e., $\operatorname{div}(\sigma)_i = \operatorname{div}(\sigma_{i,\cdot})$. If we assume that the body behaves linearly elastically and that the deformation is small, the stresses can be related to the displacement field of the body $\vec{u} : \Omega \to \mathbb{R}^3$ by the Lamé equations

$$\sigma = 2\mu\varepsilon(\vec{u}) + \lambda \operatorname{div}(\vec{u}) I.$$

These are a generalization of Hooke's law for an elastic spring. The coefficients μ and λ are the Lamé parameters, and the linearized strain tensor

$$\varepsilon(\vec{u}) = \tfrac{1}{2}\left(D\vec{u} + (D\vec{u})^\top\right)$$

describes the local stretching of the material. Here $(D\vec{u})_{ij} = \partial_j u_i$ is the Jacobian matrix of the displacement field $\vec{u}$, and hence $\varepsilon(\vec{u})$ amounts to a symmetrized gradient operator. Putting everything together, we arrive at the linear (better: linearized) elasticity equations

$$-\operatorname{div}\left(2\mu\varepsilon(\vec{u}) + \lambda \operatorname{div}(\vec{u}) I\right) = \vec{f} \qquad \text{in } \Omega. \tag{1.10}$$

This is a system of linear partial differential equations of second order. At the boundary, we may prescribe the displacement $\vec{u}$ or the normal forces, e.g., by

$$\vec{u} = \vec{g} \qquad \text{on } \Gamma_D \tag{1.11}$$

$$\vec{n} \cdot \sigma = 0 \qquad \text{on } \Gamma_N = \partial\Omega \setminus \Gamma_D. \tag{1.12}$$

Using the relation between stresses and strains, the normal stress can be expressed in terms of the displacement $\vec{u}$. The boundary value problem (1.10)–(1.12) approximately describes the deformation $\vec{u}$ of a linearly elastic body under the action of an external load. Note that the equations are an approximation that is valid only for small deformations $\vec{u}$. For large deformations, the strain tensor $\varepsilon(\vec{u})$ has to be replaced by a nonlinear expression.

With similar arguments as used in the previous examples, one can see that the system (1.10)–(1.12) amounts to the Euler-Lagrange equations (=first order optimality conditions) for the variational problem

$$\frac{1}{2}\int_\Omega 2\mu|\varepsilon(\vec{u})|^2 + \lambda|\operatorname{div}(\vec{u})|^2\,dx - \int_\Omega \vec{f}\cdot\vec{u}\,dx \to \min_{\vec{u}:\vec{u}|_{\Gamma_D}=\vec{g}}.$$

This can be verified with the same arguments as for the membrane problem; see the exercises. In the elasticity context, the test functions $\vec{v}$ appearing in this derivation are called *virtual displacements*. As we will see, the analytical and numerical tools developed for our model problem (1.4)–(1.5) will apply almost immediately also to the linear elasticity system. Further details will again be discussed in the Lecture: *Continuum Mechanics*.

Incompressible limit

For some particular materials, such as rubber, the parameter λ becomes extremely large. It is then convenient to introduce an auxiliary variable $p = -\lambda \operatorname{div}(\vec{u})$, which allows us to rewrite (1.10) as an equivalent system

$$\begin{aligned} -\operatorname{div}(2\mu\varepsilon(\vec{u})) + \nabla p &= \vec{f} \qquad \text{in } \Omega \\ \operatorname{div}(\vec{u}) + \tfrac{1}{\lambda} p &= 0 \qquad \text{in } \Omega. \end{aligned}$$

These equations now have a different structure than the elliptic problems considered before, and we will only briefly comment on the generalization of our arguments to such problems; details are discussed in the Lecture: *Numerical methods for Continuum Mechanics*. Note that the case of large λ now corresponds to a small perturbation of the limit problem, for which the second equation reads

$$\operatorname{div}(\vec{u}) = 0 \qquad \text{in } \Omega.$$

This constraint restricts the set of possible deformations of an incompressible material. Note that $\int_\omega \operatorname{div}(\vec{u})\, dx = 0$ for all $\omega \subset \Omega$, i.e., the first-order (infinitesimal) change in volume of every material subdomain vanishes. Exact volume preservation in the nonlinear theory would instead require $\det(I + D\vec{u}) = 1$.

1.5 Incompressible flow

The incompressible flow of a homogeneous Newtonian fluid is described by the famous Navier-Stokes equations

$$\begin{aligned} \partial_t \vec{u} + \vec{u} \cdot \nabla \vec{u} - \nu \Delta \vec{u} + \nabla p &= \vec{f} \\ \operatorname{div}(\vec{u}) &= 0. \end{aligned}$$

Here $\vec{u}$ denotes the flow velocity, p is the pressure, and ν the kinematic viscosity. Under steady flow conditions and assuming slow flow velocities, the first two terms in the first equation can be dropped and we arrive at the Stokes system

$$-\nu \Delta \vec{u} + \nabla p = \vec{f} \qquad \text{in } \Omega, \tag{1.13}$$

$$\operatorname{div}(\vec{u}) = 0 \qquad \text{in } \Omega. \tag{1.14}$$

These equations describe the slow, steady and incompressible flow of a homogeneous Newtonian fluid. To complete the problem description, additional boundary conditions are required, e.g.,

$$\vec{u} = 0 \qquad \text{on } \partial\Omega. \tag{1.15}$$

This condition means that the fluid perfectly sticks to the boundary and hence the velocity is zero there (=no slip condition).

Apart from the physical meaning of the variables $\vec{u}$ and p, the equations have exactly the same form as the incompressible limit of the linear elasticity system, and thus can be handled with the same analytical and numerical tools. With similar arguments as employed for the minimal surface problem, one can see, for instance, that solutions of (1.13)–(1.15) characterize the minima of

$$\frac{1}{2}\int_\Omega \nu|D\vec{u}|^2\,dx - \int_\Omega \vec{f}\cdot\vec{u}\,dx \to \min_{\substack{\vec{u}:\vec{u}|_{\partial\Omega}=0\\ \operatorname{div}(\vec{u})=0}}.$$

The steady flow of an incompressible fluid and the deformation of an incompressible elastic material are thus again characterized by a variational problem encoding the principle of minimum energy. The constrained Euler–Lagrange equation and the role of the pressure are discussed in the exercises.

1.6 Magneto-quasistatics

The propagation of electromagnetic waves is governed by Maxwell's equations. The two most important relations, due to Ampère and Faraday, are

$$\operatorname{curl}\vec{H} = \partial_t\vec{D} + \vec{J}, \qquad \operatorname{curl}\vec{E} = -\partial_t\vec{B}.$$

These are balance laws for the electric and magnetic fields $\vec{E}$, $\vec{H}$, the corresponding fluxes $\vec{D}$, $\vec{B}$, also known as electric displacement and magnetic induction, and the current $\vec{J}$. Note that all these functions are vector valued!

To obtain a closed system, one again requires additional material laws. For a linear isotropic non-dispersive medium, these are given by

$$\vec{D} = \varepsilon\vec{E}, \qquad \vec{B} = \mu\vec{H}, \qquad \vec{J} = \vec{J}_s + \sigma\vec{E},$$

with permittivity ε, permeability μ and conductivity σ. The term $\vec{J}_s$ denotes the impressed currents, which act as sources for the electromagnetic fields.

By taking the divergence in Faraday's law and using $\operatorname{div}\operatorname{curl} = 0$, one can deduce that $\operatorname{div}(\partial_t\vec{B}) = 0$. Together with appropriate initial conditions, this

implies $\operatorname{div}(\vec{B}) = 0$. On a simply connected domain, the magnetic induction can thus be expressed in the form

$$\vec{B} = \operatorname{curl} \vec{A},$$

with $\vec{A}$ denoting a magnetic vector potential. Since $\operatorname{curl} \nabla\phi = 0$, we then also have $\vec{B} = \operatorname{curl}(\vec{A} - \nabla\phi)$ for all smooth scalar functions ϕ. We here assume that ϕ is chosen, such that $\vec{E} = -\partial_t \vec{A}$, which can again be justified via Faraday's law. With the help of these transformations and the material laws stated above, the remaining Ampère law can be rewritten as

$$\varepsilon \partial_{tt} \vec{A} + \sigma \partial_t \vec{A} + \operatorname{curl}(\mu^{-1} \operatorname{curl} \vec{A}) = \vec{J}_s.$$

In the magneto-quasistatic approximation, the first term in this equation, which corresponds to $\partial_t \vec{D}$, is neglected. After Laplace-transformation in time, we obtain

$$\kappa \vec{a} + \operatorname{curl}(\mu^{-1} \operatorname{curl} \vec{a}) = \vec{j}_s \qquad \text{in } \Omega, \tag{1.16}$$

with $\kappa = s\sigma$, $\vec{a} = \mathcal{L}(\vec{A})$, and $\vec{j}_s = \mathcal{L}(\vec{J}_s)$. This is a second order elliptic partial differential equation, parametrized by the Laplace-parameter s. The corresponding homogeneous Dirichlet and Neumann boundary condition now read

$$\vec{n} \times \vec{a} = 0 \qquad \text{on } \Gamma_D \tag{1.17}$$

$$\vec{n} \times (\mu^{-1} \operatorname{curl} \vec{a}) = 0 \qquad \text{on } \partial\Omega \setminus \Gamma_D. \tag{1.18}$$

An appropriate weak formulation for this problem will be derived later.

The two-dimensional case

A setting of practical relevance, e.g. for electric machine simulation, is that $\Omega = \omega \times (0, L)$ and the magnetic and electric fields have the form $\vec{B} = (B_x, B_y, 0)$ and $\vec{E} = (0, 0, E_z)$, with component functions independent of z. We deduce that $\vec{a} = (0, 0, u)$ and $\vec{j}_s = (0, 0, f)$ with u and f depending on x and y only. In this case, the system (1.16)–(1.18) can be transformed into

$$\begin{aligned} \kappa u - \operatorname{div}(\mu^{-1} \nabla u) &= f \qquad \text{in } \omega, \\ u &= 0 \qquad \text{on } \partial\omega. \end{aligned}$$

For $\kappa = 0$, this again corresponds to our model problem (1.4)–(1.5). All results obtained for this model problem therefore apply also to the magneto-quasistatic problem in the 2-dimensional setting. As we will see later, most of the results can also be generalized to the three dimensional problem (1.16)–(1.18). Details will be discussed in the Lecture: *Computational Electromagnetics*.

1.7 Concluding remarks

All problems mentioned above are derived from similar physical principles, i.e. conservation or balance laws, constitutive equations, and similar mathematical arguments. They often lead to systems of partial differential equations with similar structure. In many cases, the differential equations, more precisely: the corresponding boundary value problems, are related to some minimization principle. These observations will allow us to develop a rather general approach to treat all these problems, analytically and numerically, within a common framework.

Variational problems

We will mainly develop our ideas for the model problem

$$\begin{aligned} -\operatorname{div}(a\nabla u) &= f \qquad \text{in } \Omega, \\ u &= 0 \qquad \text{on } \partial\Omega, \end{aligned}$$

but later show, that the concepts generalize nicely to the other problems mentioned above. We may multiply the differential equation by a test function v, integrate over the domain and use the integration-by-parts formula, to see that

$$\begin{aligned} \int_\Omega f\, v\, dx &= \int_\Omega -\operatorname{div}(a\nabla u)\, v\, dx \\ &= \int_\Omega a\nabla u \cdot \nabla v\, dx - \int_{\partial\Omega} a\partial_n u\, v\, ds. \end{aligned}$$

Here we used $\vec{n} \cdot (a\nabla u) = a\,(\vec{n} \cdot \nabla u) =: a\partial_n u$ in the boundary term. Any smooth solution of our model problem thus also satisfies the variational principle

$$\int_\Omega a\nabla u \cdot \nabla v\, dx = \int_\Omega f\, v\, dx \qquad \forall v : v|_{\partial\Omega} = 0. \tag{VP}$$

Since only first derivatives appear in this equation, we will call this the *weak* or *variational form* of our boundary value problem. Let us note that additionally the condition $u|_{\partial\Omega} = 0$ still has to be required. The weak formulation of all other model problems will have a very similar structure, which suggests to consider such problems on a more abstract level.

2 Elliptic variational problems

In this chapter, we study linear variational problems of the abstract form

$$\text{Find } u \in V: \qquad a(u,v) = \ell(v) \qquad \forall v \in V, \tag{VP}$$

where V is some Hilbert space, $a : V \times V \to \mathbb{R}$ is a bilinear form, and $\ell : V \to \mathbb{R}$ is a linear mapping. We always assume that V is a vector space over $\mathbb{R}$; the complex case can be considered with minor modifications. The scalar product and norm of V are denoted by $(\cdot,\cdot)_V$ and $\|\cdot\|_V$, and we write V^* for the dual space of V, which consists of all linear functionals $\ell : V \to \mathbb{R}$ with bounded norm

$$\|\ell\|_{V^*} := \sup_{0 \neq v \in V} \frac{|\ell(v)|}{\|v\|_V}.$$

The action of a linear functional can also be expressed as $\ell(v) = \langle \ell, v \rangle_{V^* \times V}$ via the duality product. As a consequence of these definitions, we have

$$\ell(v) = \langle \ell, v \rangle_{V^* \times V} \leq \|\ell\|_{V^*} \|v\|_V,$$

which holds for all $\ell \in V^*$ and all $v \in V$, and resembles the Cauchy-Schwarz inequality $(u,v)_V \leq \|u\|_V \|v\|_V$. The following notions will be of importance.

Definition 2.1. The bilinear form $a : V \times V \to \mathbb{R}$ is called

- *elliptic* (*coercive*), if there exists $\alpha > 0$ with $a(u,u) \geq \alpha \|u\|_V^2$ for all $u \in V$;
- *continuous*, if there is a $C_a > 0$ with $|a(u,v)| \leq C_a \|u\|_V \|v\|_V$ for all $u, v \in V$;
- *symmetric*, if $a(u,v) = a(v,u)$ for all $u, v \in V$;
- *inner product* if it is symmetric, $a(u,u) \geq 0$ for all $u \in V$ and $a(u,u) = 0$ implies $u = 0$.

Convention. We will always use the same symbols α and C_a for the ellipticity and continuity constants, when we require that $a(\cdot,\cdot)$ has one of these properties.

2.1 Symmetric variational problems

Most of the following material should be familiar from the Lectures: *Partial Differential Equations* and *Numerical methods for Differential Equations*. For completeness of the presentation, we recall the main theoretical results here.

Lemma 2.2 (Equivalent minimization problem)**.**
Let $a : V \times V \to \mathbb{R}$ be symmetric, continuous, and non-negative, i.e., $a(u,u) \geq 0$ for all $u \in V$. Then (VP) *is equivalent to the minimization problem*

$$J(u) = \min_{w \in V} J(w) \qquad \text{with } J(w) := \tfrac{1}{2}a(w,w) - \ell(w). \tag{MP}$$

Proof. Using (bi)linearity of $a(\cdot,\cdot)$ and $\ell(\cdot)$, we see that

$$J(u+tv) = J(u) + t[a(u,v) - \ell(v)] + \tfrac{t^2}{2}a(v,v) \qquad \forall t \in \mathbb{R}\ \forall v \in V.$$

($\Rightarrow$) If (VP) holds, i.e., $a(u,v) = \ell(v)$, then $J(u+tv) = J(u) + \frac{t^2}{2}a(v,v) \geq J(u)$, since $a(\cdot,\cdot)$ was assumed non-negative. Hence u is a minimizer.

($\Leftarrow$) If u is a minimizer of J, then, for every $v \in V$, the function $t \mapsto J(u+tv)$ has a minimum at $t = 0$. Hence

$$0 = \frac{d}{dt}J(u+tv)\Big|_{t=0} = a(u,v) - \ell(v).$$

Thus $a(u,v) = \ell(v)$ for all $v \in V$, and (VP) holds. □

As a next step, we show existence of a minimizer for (MP), which by the previous consideration amounts to a solution of the variational problem (VP).

Lemma 2.3 (Lax-Milgram, symmetric version)**.**
Let $a : V \times V \to \mathbb{R}$ be elliptic, continuous, and symmetric. Then for any $\ell \in V^*$, the variational problem (VP) *has a unique solution $u \in V$ and $\|u\|_V \leq \frac{1}{\alpha}\|\ell\|_{V^*}$.*

Proof. We proceed as in the proof of the Riesz representation theorem, which is the special case $a(u,v) := (u,v)_V$; see the Lecture: *Functional Analysis*.

(i) From the properties of $a(\cdot,\cdot)$ and $\ell(\cdot)$, we see that

$$\begin{aligned} J(u) &= \tfrac{1}{2}a(u,u) - \ell(u) \geq \tfrac{\alpha}{2}\|u\|_V^2 - \|\ell\|_{V^*}\|u\|_V \\ &\geq \tfrac{\alpha}{2}(\|u\|_V - \tfrac{1}{\alpha}\|\ell\|_{V^*})^2 - \tfrac{1}{2\alpha}\|\ell\|_{V^*}^2 \geq -\tfrac{1}{2\alpha}\|\ell\|_{V^*}^2. \end{aligned}$$

This shows that $J(\cdot)$ is bounded from below and hence there exists $c := \inf_{v \in V} J(v)$.

(ii) By definition of the infimum, we can find a (minimizing) sequence $(u_n)_{n\in\mathbb{N}}$ such that $c \leq J(u_n) \leq c + \frac{1}{n}$. Using ellipticity and bilinearity of a, we then get

$$\begin{aligned} \alpha\|u_n - u_m\|_V^2 &\leq a(u_n - u_m, u_n - u_m) \\ &\overset{(*)}{=} 2a(u_n,u_n) + 2a(u_m,u_m) - a(u_n+u_m, u_n+u_m) \\ &= 4J(u_n) + 4J(u_m) - 8J(\tfrac{u_n+u_m}{2}) \\ &\leq 4(c+\tfrac{1}{n}) + 4(c+\tfrac{1}{m}) - 8c = 4(\tfrac{1}{n} + \tfrac{1}{m}). \end{aligned}$$

This shows that $(u_n)_{n\in\mathbb{N}}$ is a Cauchy sequence. The identity (*) used in the second step is called *parallelogram identity*.

(iii) Since V is a Hilbert space, and hence complete, there exists a limit $u := \lim_n u_n$, and by continuity of the functional $J(u)$, we see that

$$c = \lim_{n\to\infty} J(u_n) = J(\lim_{n\to\infty} u_n) = J(u),$$

which shows that u is a minimizer. By equivalence of the minimization (MP) with the variational problem, we see that u also is a solution of (VP).

(iv) Using ellipticity of $a(\cdot,\cdot)$ and continuity of $\ell(\cdot)$, we further see that

$$\alpha\|u\|_V^2 \le a(u,u) = \ell(u) \le \|\ell\|_{V^*}\|u\|_V,$$

which yields the a-priori bound. By linearity of the problem (VP), the difference $w = u_1 - u_2$ of two potential solutions satisfies (VP) with $\ell = 0$, and hence the solution is unique; see the exercises. □

Let us discuss an immediate consequence of this result.

Corollary 2.4 (Riesz representation theorem)**.**
For any $\ell \in V^*$, there exists a unique $u_\ell \in V$ such that $\ell = (u_\ell, \cdot)_V$, i.e.,

$$(u_\ell, v)_V = \ell(v) = \langle \ell, v\rangle_{V^*\times V} \qquad \forall v \in V.$$

Moreover, the mapping $R_V : V^* \to V$, $\ell \mapsto u_\ell$, is linear, continuous, and continuously invertible with $\|R_V\ell\|_V = \|\ell\|_{V^*}$ for all $\ell \in V^*$ and is called the Riesz isomorphism.

For later reference, let us mention another direct consequence of Lemma 2.3.

Corollary 2.5 (Projection lemma)**.**
Let $W \subset V$ be a closed subspace of the Hilbert space V. Then for any $u \in V$ there exists a unique (closest point) element $u_W \in W$ such that

$$\|u - u_W\|_V = \min_{w\in W} \|u - w\|_V. \tag{2.1}$$

Moreover, the minimizer is characterized by

$$(u - u_W, w)_V = 0 \qquad \forall w \in W. \tag{2.2}$$

Proof. Since W is closed, it is a Hilbert space with the scalar product inherited from V. Applying Lemma 2.3 on W to

$$a(w,z) := (w,z)_V, \qquad \ell(z) := (u,z)_V,$$

gives a unique $u_W \in W$ such that

$$(u - u_W, z)_V = 0 \qquad \forall z \in W.$$

For arbitrary $w \in W$, we have $u_W - w \in W$, and thus

$$\begin{aligned} \|u-w\|_V^2 &= \|u - u_W + (u_W - w)\|_V^2 \\ &= \|u-u_W\|_V^2 + 2(u-u_W, u_W - w)_V + \|u_W - w\|_V^2 \\ &= \|u-u_W\|_V^2 + \|u_W - w\|_V^2 \geq \|u-u_W\|_V^2. \end{aligned}$$

Thus u_W is the unique minimizer and is characterized by (2.2). □

Remark 2.6. The element $u_W \in W$ in the above lemma is the best-approximation for $u \in V$ in the closed subspace $W \subset V$. The variational equation (2.2) states that the approximation error $u - u_W$ is orthogonal to the approximating subspace W; draw a sketch! This is a special instance of *Galerkin orthogonality*, discussed in more detail later on. Taking $w = 0$ in the identity proved above gives

$$\|u\|_V^2 = \|u_W\|_V^2 + \|u - u_W\|_V^2.$$

Any function $u \in V$ can thus be split orthogonally into $u = u_W + (u - u_W)$ with $u_W \in W$ and $u - u_W \in W^\perp$, which is the orthogonal complement of W in V.

2.2 The lemma of Lax-Milgram

A main observation of Peter Lax and Arthur Milgram was that the symmetry of the bilinear form is not required to obtain a unique solution of (VP).

Lemma 2.7 (Lax-Milgram, 1954)**.**
Let $a : V \times V \to \mathbb{R}$ be an elliptic and continuous bilinear form. Then for any $\ell \in V^*$, the problem (VP) *has a unique solution $u \in V$ and $\|u\|_V \leq \frac{1}{\alpha}\|\ell\|_{V^*}$.*

Proof. The mapping $A : V \to V^*$, $u \mapsto a(u,\cdot)$, is linear and bounded with $\|Au\|_{V^*} \leq C_a\|u\|_V$. The variational problem (VP) is equivalent to the operator equation (see exercises)

$$Au = \ell \qquad \text{in } V^*. \tag{OP}$$

With the Riesz isomorphism R_V we write (OP) as the fixed-point problem

$$u = u + \omega R_V(\ell - Au) =: \Phi_\omega(u) \qquad \text{in } V. \tag{FP}$$

We choose $\omega = \alpha/C_a^2$. For $z := u - v$, ellipticity and continuity yield

$$\begin{aligned}\|\Phi_\omega(u) - \Phi_\omega(v)\|_V^2 &= \|z - \omega R_V A z\|_V^2 \\ &= \|z\|_V^2 - 2\omega\langle Az, z\rangle_{V^*\times V} + \omega^2\|Az\|_{V^*}^2 \\ &\le \big(1 - 2\alpha\omega + C_a^2\omega^2\big)\|z\|_V^2 = \left(1 - \frac{\alpha^2}{C_a^2}\right)\|z\|_V^2.\end{aligned}$$

If $V = \{0\}$, the assertion is trivial. Otherwise, ellipticity and continuity imply $0 < \alpha \le C_a$, so Φ_ω is a contraction. Banach's fixed-point theorem therefore gives a unique solution of (FP), and hence of (OP) and (VP). Finally, using $v = u$ in (VP) gives

$$\alpha\|u\|_V^2 \le a(u,u) = \ell(u) \le \|\ell\|_{V^*}\|u\|_V,$$

and thus $\|u\|_V \le \alpha^{-1}\|\ell\|_{V^*}$. □

2.3 Galerkin approximation and Céa's lemma

We next discuss the systematic approximation of elliptic variational problems.

Problem 2.8 (Galerkin approximation, 1915)**.**
Let $V_h \subset V$ be some closed (e.g. finite-dimensional) subspace.

$$\text{Find } u_h \in V_h\text{:} \qquad a(u_h, v_h) = \ell(v_h) \qquad \forall v_h \in V_h. \tag{VP$_h$}$$

Under the assumptions of the Lax-Milgram lemma, we immediately obtain a unique *discrete* solution $u_h \in V_h$ and $\|u_h\|_V \le \frac{1}{\alpha}\|\ell\|_{V^*}$.

Remark 2.9 (Galerkin-isomorphism)**.**
Let $\dim(V_h) = N$ and $\{\phi_1, \dots, \phi_N\} \subset V_h$ denote a basis of V_h. Then any element $u_h \in V_h$ can be expanded in this basis and

$$G_h : \mathbb{R}^N \to V_h, \qquad \underline{u} \mapsto \sum\nolimits_i \underline{u}_i\phi_i$$

is a linear and bijective mapping, the so-called *Galerkin isomorphism*. Moreover, the variational problem (VP$_h$) is equivalent to the linear system

$$\underline{A}\,\underline{u} = \underline{b} \qquad \text{in } \mathbb{R}^N, \tag{2.3}$$

with $\underline{A}_{ij} = a(\phi_j, \phi_i)$ and $\underline{b}_i = \ell(\phi_i)$; see the exercises and compare with (OP) in the proof of the Lax-Milgram lemma. From the ellipticity of $a(\cdot,\cdot)$, one can further deduce that $\underline{A}$ is positive definite, and hence regular.

We thus know that the Galerkin approximation u_h to u is well-defined and, in principle, even how to compute it. A natural next question is, of course, how well it approximates the true solution. This is answered by the following result.

Lemma 2.10 (Jean Céa, 1967, PhD thesis)**.**
Let $a : V \times V \to \mathbb{R}$ be elliptic and continuous. Then

$$\|u - u_h\|_V \leq \frac{C_a}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|_V.$$

Proof. Subtracting both variational equations yields the Galerkin orthogonality

$$a(u - u_h, w_h) = a(u, w_h) - a(u_h, w_h) = \ell(w_h) - \ell(w_h) = 0 \qquad \forall w_h \in V_h. \quad \text{(GO)}$$

Let $v_h \in V_h$ be arbitrary and set $e := u - u_h$. Since $v_h - u_h \in V_h$, ellipticity, Galerkin orthogonality, and continuity give

$$\begin{aligned} \alpha\|e\|_V^2 &\leq a(e,e) = a(e, u - v_h) + a(e, v_h - u_h) \\ &= a(e, u - v_h) \leq C_a \|e\|_V \|u - v_h\|_V. \end{aligned}$$

If $e = 0$, the assertion is immediate. Otherwise, division by $\|e\|_V$ yields

$$\|u - u_h\|_V \leq \frac{C_a}{\alpha} \|u - v_h\|_V.$$

Taking the infimum over all $v_h \in V_h$ finishes the proof. □

The symmetric case and the Ritz method

If $a(\cdot,\cdot)$ is also symmetric, then the discrete variational problem (VP_h) is equivalent to the discrete minimization problem

$$\frac{1}{2} a(u_h, u_h) - \ell(u_h) \to \min_{u_h \in V_h}.$$

Using the Galerkin isomorphism $u_h = \sum_i \underline{u}_i \phi_i$, this can also be phrased as

$$\min_{\underline{u} \in \mathbb{R}^N} \frac{1}{2} \underline{u}^\top \underline{A}\underline{u} - \underline{u}^\top \underline{b},$$

i.e., as a quadratic minimization problem in $\mathbb{R}^N$. Standard optimization algorithms, e.g. the conjugate gradient method, can be used for its solution.

Walter Ritz used such problems before 1910 to compute approximate solutions (by hand!) for the biharmonic equation and similar variational problems. In the symmetric case, the bilinear form $a(u, v)$ defines a scalar product on V, and $\|u\|_a = \sqrt{a(u, u)}$ is the corresponding norm. From the projection lemma, we then immediately obtain

$$\|u - u_h\|_a \leq \|u - v_h\|_a \qquad \forall v_h \in V_h.$$

In the symmetric case, the Galerkin approximation u_h is thus the best approximation to u in the space V_h with respect to the energy norm. Together with the ellipticity and continuity of the bilinear form, this yields

$$\|u - u_h\|_V \leq \sqrt{\frac{C_a}{\alpha}} \inf_{v_h \in V_h} \|u - v_h\|_V,$$

which is somewhat tighter than the estimate from Céa's lemma; see the exercises.

2.4 Inf-sup stable variational problems

Ellipticity is sufficient for well-posedness, but it is not necessary. We now allow possibly different real Hilbert spaces V and W for trial and test elements. We write W^* for the space of bounded linear functionals on W, with norm $\|F\|_{W^*} := \sup_{0 \neq v \in W} |F(v)|/\|v\|_W$. Let $b : V \times W \to \mathbb{R}$ be a continuous bilinear form, with $|b(u, v)| \leq C_b \|u\|_V \|v\|_W$, and define $B : V \to W^*$ by $(Bu)(v) := b(u, v)$. For $\ell \in W^*$ we seek $u \in V$ such that $Bu = \ell$.

Lemma 2.11 (Inf-sup criterion)**.**
Suppose that

$$\inf_{0 \neq u \in V} \sup_{0 \neq v \in W} \frac{|b(u, v)|}{\|u\|_V \|v\|_W} \geq \beta > 0,$$

and that for every $0 \neq v \in W$ there is a $u \in V$ with $b(u, v) \neq 0$. Then for every $\ell \in W^*$ there is a unique solution of $Bu = \ell$, and $\|u\|_V \leq \beta^{-1} \|\ell\|_{W^*}$.

The inf-sup bound says $\|Bu\|_{W^*} \geq \beta \|u\|_V$: it makes B injective and its range closed. Let $R_W : W^* \to W$ be the Riesz isomorphism, so $b(u, v) = (R_W Bu, v)_W$. The second condition means that $R_W(B(V))^\perp = \{0\}$. Since $R_W(B(V))$ is closed,

it equals W, and hence $B(V) = W^*$. A sufficient condition for the second assumption is the *reversed inf-sup bound*

$$\inf_{0\neq v\in W} \sup_{0\neq u\in V} \frac{|b(u,v)|}{\|u\|_V \|v\|_W} > 0.$$

This is the quantitative surjectivity counterpart to the first bound.

Example 2.12. On $V = W = \mathbb{R}^2$ with the Euclidean norm, let $b(u,v) = u_1v_1 - u_2v_2$. Choosing $v = (u_1, -u_2)$ gives $b(u,v) = \|u\|^2$ and $\|v\| = \|u\|$, and Cauchy–Schwarz gives the upper bound. The same argument applies with u and v interchanged, so both inf-sup conditions hold. Yet $b((1,1),(1,1)) = 0$, so b is not elliptic.

For finite-dimensional $V_h \subset V$ and $W_h \subset W$ with $\dim V_h = \dim W_h$, we seek $u_h \in V_h$ such that $b(u_h, v_h) = \ell(v_h)$ for all $v_h \in W_h$. The continuous inf-sup bound does not need to hold on $V_h \times W_h$. In the example above, choosing $V_h = W_h = \operatorname{span}\{(1,1)\}$ makes b vanish identically on $V_h \times W_h$. Therefore the discrete problem requires its own bound

$$\inf_{0\neq u_h\in V_h} \sup_{0\neq v_h\in W_h} \frac{|b(u_h,v_h)|}{\|u_h\|_V \|v_h\|_W} \geq \beta_h > 0.$$

Equal dimensions and this bound give a unique discrete solution. For a family of such pairs (V_h, W_h) indexed by h, uniform stability and error estimates require $\beta_h \geq \beta_0 > 0$ with β_0 independent of h. Ellipticity, by contrast, is inherited by every conforming Galerkin discretization.

Remark 2.13 (Exercise)**.** The closed-range and surjectivity arguments, as well as the discrete Céa-type estimate, are developed in the exercises.

2.5 Density argument

Before closing this section, let us state an auxiliary result, which will be used intensively in the following chapter. Recall that a subspace $S \subset V$ is called *dense* in V if any element $u \in V$ can be approximated well by elements in S, i.e.,

$$\forall u \in V\ \forall n \in \mathbb{N}\ \exists s_n \in S : \|u - s_n\|_V \leq 1/n.$$

A linear operator $A : V \to W$ between normed spaces is called bounded if

$$\|A\|_{V\to W} := \sup_{0\neq v\in V} \frac{\|Av\|_W}{\|v\|_V} < \infty.$$

We then immediately obtain the following result.

Lemma 2.14 (Closure principle, density argument)**.**
Let $A : S \subset V \to W$ be a densely defined bounded linear operator between Hilbert (or Banach) spaces V, W. Then there exists a unique bounded linear operator $\bar{A} : V \to W$, called the extension *or* closure *of A, with $\bar{A}s = As$ for all $s \in S$. Moreover, $\|\bar{A}\|_{V\to W} = \|A\|_{V\to W}$.*

Proof. Let $u \in V$ be arbitrary and $(s_n)_{n\in\mathbb{N}} \subset S$ such that $\|u - s_n\|_V \leq 1/n$. Since A is linear and bounded, we see that As_n is Cauchy in the complete space W

$$\|As_n - As_m\|_W \leq \|A\|_{V\to W}\|s_n - s_m\|_V.$$

We may thus set $\bar{A}u := \lim_n As_n$. The same argument shows that the value is independent of the approximating sequence. The extension property, linearity, norm bound, and uniqueness are left as an exercise. □

3 Sobolev spaces

In the following, we introduce the function spaces that are of relevance for the weak formulation of elliptic partial differential equations. Details can be found in Evans: *Partial differential equations* and the Lecture: *Functional Analysis*.

Preliminaries and notation

Throughout this section, we let $\Omega \subset \mathbb{R}^d$ denote some bounded *domain*, i.e., a bounded, *open and connected set*. We denote by $L^p(\Omega)$, $1 \leq p \leq \infty$, the Lebesgue spaces of measurable functions on Ω with bounded norm

$$\|u\|_{L^p(\Omega)} = \left(\int_\Omega |u|^p dx \right)^{1/p}, \qquad 1 \leq p < \infty,$$

and $\|u\|_{L^\infty(\Omega)} = \operatorname{ess\,sup}_{x\in\Omega} |u(x)|$. Let us recall that $L^p(\Omega)$ actually consists of equivalence classes of functions that only differ on sets of measure zero and hence cannot be distinguished by the norms given above.

For all $1 \leq p \leq \infty$, the spaces $L^p(\Omega)$ are complete and $L^2(\Omega)$ is a Hilbert space with scalar product $(u, v)_{L^2(\Omega)} = \int_\Omega uv\, dx$. For $p < \infty$, the spaces $L^p(\Omega)$ are separable, i.e., they contain a countable dense set. Also recall Hölder's inequality

$$\left| \int_\Omega uv\, dx \right| \leq \|u\|_{L^p} \|v\|_{L^q}, \qquad \tfrac{1}{p} + \tfrac{1}{q} = 1,$$

which holds for all functions $u \in L^p(\Omega)$ and $v \in L^q(\Omega)$ with the corresponding indices. For $p = q = 2$, we obtain the Cauchy-Schwarz inequality.

3.1 Weak derivatives and Sobolev spaces

We denote by $C_0^\infty(\Omega)$ the set of infinitely differentiable functions with compact support, i.e., functions ϕ that vanish outside a compact set $K_\phi \subset \Omega$. Since the domain Ω was assumed open, any function $\phi \in C_0^\infty(\Omega)$ thus vanishes identically, i.e., together with all its derivatives, in the vicinity of the boundary $\partial\Omega$.

Distributional and weak derivatives

For smooth functions u and multi-index $\alpha \in \mathbb{N}_0^d$, we denote by $D^\alpha u = \partial_1^{\alpha_1} \cdots \partial_d^{\alpha_d} u$ the classical derivatives of order α. Using integration-by-parts, we see that

$$\int_\Omega D^\alpha u \, \phi \, dx = (-1)^{|\alpha|} \int_\Omega u \, D^\alpha \phi \, dx \qquad \forall \phi \in C_0^\infty(\Omega).$$

All boundary terms vanish here, since ϕ has compact support. The right hand side of this formula is well-defined for any locally integrable function u and can be used to define the notion of a derivative for this rather large class of functions.

Definition 3.1 (Distributional and weak derivative)**.**
(i) Let $L^1_{\mathrm{loc}}(\Omega)$ denote the set of locally integrable functions, i.e., all measurable functions u with $u|_K \in L^1(K)$ for all compact subsets $K \subset \Omega$. Then for any function $u \in L^1_{\mathrm{loc}}(\Omega)$, the linear mapping $T_u^\alpha : C_0^\infty(\Omega) \to \mathbb{R}$ defined by

$$\langle T_u^\alpha, \phi \rangle := (-1)^{|\alpha|} \int_\Omega u \, D^\alpha \phi \, dx, \qquad \phi \in C_0^\infty(\Omega)$$

is called the *distributional derivative* (of index α) of the function u.

(ii) If there exists a function $g^\alpha \in L^1_{\mathrm{loc}}(\Omega)$ such that T_u^α can be expressed as

$$\langle T_u^\alpha, \phi \rangle = \int_\Omega g^\alpha \, \phi \, dx \qquad \forall \phi \in C_0^\infty(\Omega),$$

then T_u^α is called *regular*, and g^α is called the *weak derivative* of u. If all weak derivatives for $|\alpha| \le k$ exist, then u is called k-times *weakly differentiable*.

Remark 3.2. (i) Distributional derivatives can be defined for an even larger class of so-called *generalized functions*, or distributions; see Lecture: PDEs. Any function $u \in L^1_{\mathrm{loc}}(\Omega)$ can again be identified with a *regular* distribution. For any distribution, derivatives of arbitrary order always exist in *distributional sense*.

(ii) A function $u \in C^k(\Omega)$ is also k-times weakly differentiable and the weak derivatives coincide with the usual derivatives. We therefore use the symbol $D^\alpha u$ also for the weak derivatives. Piecewise smooth functions with certain global regularity are also weakly differentiable and the weak derivative coincides with the piecewise classical derivative. Piecewise smooth functions with a genuine jump across an interface, on the other hand, are not weakly differentiable. Changes of a representative on a set of measure zero do not affect weak differentiability. In a similar manner, we can define weak versions of ∇u, $\operatorname{div} \vec{j}$, $\operatorname{curl} \vec{j}$, and so on; see the exercises.

Definition and norms

For any integer $k \geq 0$ and $1 \leq p \leq \infty$, we can now define the Sobolev spaces

$$W^{k,p}(\Omega) := \{u \in L^p(\Omega) : D^\alpha u \in L^p(\Omega) \quad \forall \alpha : |\alpha| \leq k\}.$$

These are complete, i.e., Banach spaces, when equipped with the norms

$$\|u\|_{W^{k,p}(\Omega)} = \left(\sum\nolimits_{|\alpha| \leq k} \|D^\alpha u\|^p_{L^p(\Omega)} \right)^{1/p}, \qquad 1 \leq p < \infty,$$

and $\|u\|_{W^{k,\infty}(\Omega)} = \max_{|\alpha| \leq k} \|D^\alpha u\|_{L^\infty(\Omega)}$ in the special case $p = \infty$, respectively. For $p = 2$, the Sobolev spaces are again Hilbert spaces and to emphasize this, we write $H^k(\Omega) = W^{k,2}(\Omega)$. We will also make use of the semi-norms

$$|u|_{W^{k,p}(\Omega)} = \left(\sum\nolimits_{|\alpha| = k} \|D^\alpha u\|^p_{L^p(\Omega)} \right)^{1/p}, \qquad 1 \leq p < \infty,$$

For $p = \infty$, we set $|u|_{W^{k,\infty}(\Omega)} = \max_{|\alpha|=k} \|D^\alpha u\|_{L^\infty(\Omega)}$. These seminorms take into account only the derivatives of highest order. For further details, see Evans: *Partial Differential Equations* or the Lecture: *Functional Analysis*.

3.2 Density, traces and embeddings

By construction, the Sobolev spaces again consist of equivalence classes of functions that differ only on a set of measure zero, and it seems difficult to work with these objects. The following properties allow us to deal with this issue and to calculate with elements of Sobolev spaces as with ordinary functions.

Lemma 3.3 (Density)**.**
Let $1 \leq p < \infty$ and $m \geq k$. Then the following assertions hold:

(i) If $\Omega \subset \mathbb{R}^d$ is open, then the spaces $C^m(\Omega) \cap W^{k,p}(\Omega)$ are dense in $W^{k,p}(\Omega)$.

(ii) If Ω is a bounded Lipschitz domain, then even $C^m(\overline{\Omega}) \subset W^{k,p}(\Omega)$ is dense.

The proof works by constructing an approximating sequence $\{\phi_n\}_{n \in \mathbb{N}} \subset C^\infty(\Omega)$ via so-called *mollification*; see Evans: *Partial Differential Equations* for details.

Definition 3.4. A domain $\Omega \subset \mathbb{R}^d$ is called a Lipschitz domain if there exists

(a) a finite collection $(U_n)_{n \leq N}$ of open sets, such that $\partial\Omega \subset \bigcup_{n \leq N} U_n$;

(b) bi-Lipschitz functions $f_n : O_n \subset \mathbb{R}^d \to U_n$ (Lipschitz, bijective, with Lipschitz inverse) such that $\partial\Omega \cap U_n = f_n(O_n \cap \{x_d = 0\})$ and $\Omega \cap U_n = f_n(O_n \cap \{x_d < 0\})$.

Make a sketch to visualize the meaning of this definition!

Trace theorem

As a first application of the density argument, we show that one can make sense of boundary values of a Sobolev function.

Lemma 3.5 (Trace theorem)**.**
Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain and $1 \leq p < \infty$. Then there exists a unique bounded linear operator $\gamma : W^{1,p}(\Omega) \to L^p(\partial\Omega)$, such that $\gamma u = u|_{\partial\Omega}$ for all $u \in C^1(\overline{\Omega})$ and

$$\|\gamma u\|_{L^p(\partial\Omega)} \leq C_{\mathrm{tr}} \|u\|_{W^{1,p}(\Omega)} \qquad \forall u \in W^{1,p}(\Omega)$$

with C_{tr} depending only on p and the Lipschitz character and domain size.

Proof. To illustrate the basic arguments, we sketch the proof for the special case $p = 2$ and $\Omega = (a, b) \times (c, d)$. The calculation is completed in the exercises; the general case follows by decomposition and mapping of the boundary. We denote the top part of the boundary by $\Gamma_{\mathrm{top}} := (a, b) \times \{d\}$. Since $C^1(\overline{\Omega}) \subset H^1(\Omega)$ is dense, the trace operator γ is densely defined and linear. We next show the uniform bound for $u \in C^1(\overline{\Omega})$. By the fundamental theorem of calculus, we have

$$u(x, d) = u(x, y) + \int_y^d \partial_y u(x, \eta)\, d\eta.$$

Squaring and application of the Young[1] and Cauchy-Schwarz[2] inequalities yields

$$\begin{aligned} |u(x, d)|^2 &\leq 2|u(x, y)|^2 + 2|d - y| \int_y^d |\partial_y u(x, \eta)|^2 d\eta \\ &\leq 2|u(x, y)|^2 + 2|d - c| \int_c^d |\partial_y u(x, \eta)|^2 d\eta. \end{aligned}$$

By integrating over x and y on both sides, we then arrive at

$$\begin{aligned} |d - c| \int_a^b |u(x, d)|^2 dx &\leq 2 \int_c^d \int_a^b \left[|u(x, y)|^2 + |d - c| \int_c^d |\partial_y u(x, \eta)|^2 d\eta \right] dx\, dy \\ &= 2\|u\|^2_{L^2(\Omega)} + 2|d - c|^2 \|\partial_y u\|^2_{L^2(\Omega)}. \end{aligned}$$

Hence we have shown that

$$\|u\|^2_{L^2(\Gamma_{\mathrm{top}})} \leq \tfrac{2}{|d-c|} \|u\|^2_{L^2(\Omega)} + 2|d - c|\, \|\partial_y u\|^2_{L^2(\Omega)} \leq C^2_{\mathrm{tr}} \|u\|^2_{H^1(\Omega)}.$$

[1] $ab \leq \frac{a^2}{2} + \frac{b^2}{2}$, or more general $ab \leq \varepsilon \frac{a^2}{2} + \frac{b^2}{2\varepsilon}$ for any $\varepsilon > 0$.
[2] $\|u\|_{L^1(\Omega)} = \int_\Omega |1 \cdot u|\, dx \leq \sqrt{|\Omega|}\, \|u\|_{L^2(\Omega)}$

In a similar manner, we can proceed with the other parts of the boundary, which leads to the desired trace estimate for smooth functions. Hence γ is a densely defined, linear, and bounded operator on $C^1(\overline{\Omega}) \subset H^1(\Omega)$ and can thus be extended to a bounded linear operator defined on the whole space $H^1(\Omega)$. □

The proof of the trace estimate is constructive and the constant C_{tr} only depends on the Lipschitz character and size of the domain Ω. Let us further note that the trace estimate is not sharp, i.e., not every function $g \in L^p(\partial\Omega)$ is the trace of a function $u \in W^{1,p}(\Omega)$ for $1 < p < \infty$. For details, see the literature and the next chapter.

Lemma 3.6 (Gluing of piecewise Sobolev functions)**.**
Let $\Omega_h = \{\Omega_1, \dots, \Omega_N\}$ be a partition of Ω into Lipschitz subdomains, i.e.,

$$\overline{\Omega} = \bigcup_{i=1}^{N} \overline{\Omega}_i, \qquad \Omega_i \cap \Omega_j = \emptyset \quad \text{for } i \neq j.$$

We assume that every interior intersection $\Gamma_{ij} := \partial\Omega_i \cap \partial\Omega_j \cap \Omega$, $i \neq j$, is, up to a set of zero surface measure, either empty or a Lipschitz hypersurface, with outer normal vectors $\vec{n}_j = -\vec{n}_i$ almost everywhere on its hypersurface part. Let $1 \leq p < \infty$ and $v \in L^p(\Omega)$ satisfy

$$v_i := v|_{\Omega_i} \in W^{1,p}(\Omega_i), \qquad \gamma_i v_i = \gamma_j v_j \quad \text{a.e. on } \Gamma_{ij},$$

where γ_i denotes the trace from Ω_i. Then $v \in W^{1,p}(\Omega)$ with its weak gradient $(\nabla v)|_{\Omega_i} = \nabla v_i$, $1 \leq i \leq N$.

Proof. Define $\vec{g} \in L^p(\Omega)^d$ by $\vec{g}|_{\Omega_i} := \nabla v_i$. For arbitrary $\vec{\psi} \in C_0^\infty(\Omega)^d$, integration by parts on every subdomain gives

$$\int_\Omega \vec{g} \cdot \vec{\psi}\, dx = \sum_{i=1}^{N} \int_{\Omega_i} \nabla v_i \cdot \vec{\psi}\, dx = \sum_{i=1}^{N} \left(- \int_{\Omega_i} v_i \operatorname{div}(\vec{\psi})\, dx + \int_{\partial\Omega_i} \gamma_i v_i\, \vec{\psi} \cdot \vec{n}_i\, ds \right).$$

This formula follows from the classical formula by density of $C^1(\overline{\Omega}_i)$ and continuity of the trace operator. The contributions on $\partial\Omega$ vanish because $\vec{\psi}$ has compact support in Ω. On every interface of positive surface measure, the outer normals are opposite and the traces of v_i and v_j agree. Hence they cancel and

$$\int_\Omega \vec{g} \cdot \vec{\psi}\, dx = - \int_\Omega v \operatorname{div}(\vec{\psi})\, dx.$$

Thus $\vec{g}$ is the weak gradient of v, which proves the claim. □

Remark 3.7. For piecewise C^1 functions, continuity across the interfaces implies equality of the traces. In particular, globally continuous, piecewise polynomial functions belong to $H^1(\Omega)$. Higher-order $W^{k,p}$ versions require matching traces of all derivatives up to order $k-1$.

Embedding and extension

The trace lemma is, in fact, a special case of a so-called *embedding* result, which shows that differentiability always leads to some extra integrability; the reverse is not true! For later reference, we mention the most important results, whose proof can be found in the literature.

Lemma 3.8 (Embeddings, Sobolev, Rellich-Kondrachov)**.**
Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain, $k \geq 1$, and $1 \leq p \leq \infty$. Then the embeddings

$$W^{k,p}(\Omega) \hookrightarrow \begin{cases} L^q(\Omega), & \text{for } p < \frac{d}{k} \text{ and } p \leq q \leq \frac{dp}{d-kp}, \\ L^q(\Omega), & \text{for } p = \frac{d}{k} \text{ and } p \leq q < \infty, \\ C(\overline{\Omega}), & \text{if } p > \frac{d}{k} \end{cases}$$

are continuous, i.e., $\|u\|_* \leq C_* \|u\|_{W^{k,p}(\Omega)}$ with appropriate constants C_* depending on the spaces and their canonical norms, but independent of u. Moreover

$$W^{k,p}(\Omega) \hookrightarrow \begin{cases} L^q(\Omega), & \text{for } p < \frac{d}{k} \text{ and } p \leq q < \frac{dp}{d-kp}, \\ L^q(\Omega), & \text{for } p = \frac{d}{k} \text{ and } p \leq q < \infty, \\ C(\overline{\Omega}), & \text{if } p > \frac{d}{k} \end{cases}$$

are compact and, in particular, $W^{k,p}(\Omega) \hookrightarrow W^{k-1,p}(\Omega)$ is compact.

By applying the results to $u = D^\alpha w$, we can formulate similar embedding results for $W^{k,p} \hookrightarrow W^{j,q}$ and $W^{k,p} \hookrightarrow C^j$ with $j < k$; see the exercises and the literature. For later reference, let us also mention the following useful result.

Lemma 3.9 (Extension)**.**
Let $\Omega, \widetilde{\Omega} \subset \mathbb{R}^d$ be bounded Lipschitz domains with $\overline{\Omega} \subset \widetilde{\Omega}$. Then there exists a bounded linear operator $E : W^{k,p}(\Omega) \to W^{k,p}(\widetilde{\Omega})$, $u \mapsto \widetilde{u}$ such that $\widetilde{u}|_\Omega = u$. Moreover, $\|\widetilde{u}\|_{W^{k,p}(\widetilde{\Omega})} \leq C \|u\|_{W^{k,p}(\Omega)}$ with C depending only on Ω, $\widetilde{\Omega}$, k and p.

Without going into details, let us mention that the proofs of both results, continuous embedding and extension, are constructive, i.e., the constants C_* and C can actually be estimated from properties of the domain.

Calculus in Sobolev spaces

The density lemma implies that Sobolev functions can be approximated with arbitrary accuracy in the norm of the Sobolev space by smooth functions, i.e.,

$$W^{k,p}(\Omega) = \overline{C^m(\Omega) \cap W^{k,p}(\Omega)}^{\|\cdot\|_{W^{k,p}}} \quad \text{and} \quad W^{k,p}(\Omega) = \overline{C^m(\overline{\Omega})}^{\|\cdot\|_{W^{k,p}}}, \quad 1 \le p < \infty,$$

if Ω is Lipschitz and $m \ge k$. This allows us to transfer many of the known formulas for smooth functions to Sobolev spaces, e.g., the integration-by-parts formula, the chain rule, etc., as long as all terms arising in the formulas are meaningful. Some examples are discussed in the exercises.

3.3 Equivalent Sobolev norms and Friedrichs–Poincaré inequalities

The highest-order derivatives of a Sobolev function determine it only up to a polynomial. The following compactness argument shows how additional measurements of that polynomial yield an equivalent norm. Its applications include the Friedrichs and Poincaré inequalities.

Lemma 3.10 (Polynomial kernel)**.**
Let $\Omega \subset \mathbb{R}^d$ be a bounded domain, $k \ge 1$ an integer, and $1 \le p < \infty$. If $u \in W^{k,p}(\Omega)$ satisfies $D^\alpha u = 0$ for all $|\alpha| = k$, then u agrees almost everywhere with a polynomial in $P_{k-1} := \operatorname{span}\{x^\alpha : |\alpha| \le k-1\}$.

Proof. Let $B \Subset \Omega$ (that is, $\overline{B} \subset \Omega$) be a ball and let u_ε be a standard mollification of u on B. For sufficiently small ε, differentiation commutes with mollification, so $D^\alpha u_\varepsilon = 0$ on B whenever $|\alpha| = k$. Fix $x_0 \in B$. For any $x \in B$, convexity ensures that $x_0 + t(x - x_0) \in B$ for $0 \le t \le 1$. Define $f(t) := u_\varepsilon(x_0 + t(x - x_0))$. Differentiation yields

$$f^{(k)}(t) = \sum_{|\alpha|=k} \frac{k!}{\alpha!}(x - x_0)^\alpha D^\alpha u_\varepsilon(x_0 + t(x - x_0)) = 0, \qquad \alpha! := \alpha_1! \cdots \alpha_d!.$$

The one-dimensional Taylor formula therefore has no remainder and yields

$$u_\varepsilon(x) = \sum_{j=0}^{k-1} \frac{f^{(j)}(0)}{j!} = \sum_{|\alpha| \le k-1} \frac{D^\alpha u_\varepsilon(x_0)}{\alpha!}(x - x_0)^\alpha.$$

Thus u_ε is a polynomial of degree at most $k-1$ on B. As $u_\varepsilon \to u$ in $L^p(B)$ and the finite-dimensional polynomial space is closed in $L^p(B)$, $u|_B$ is a polynomial. On

overlapping balls, these polynomials agree almost everywhere on their intersection and hence identically. Connectedness of Ω then finishes the proof. $\square$

Lemma 3.11 (Sobolev norm-equivalence theorem)**.**
Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain, $k \geq 1$ an integer, $1 \leq p < \infty$, and Z a normed space. Suppose that $T : W^{k,p}(\Omega) \to Z$ is bounded and linear and that

$$Tq = 0, \quad q \in P_{k-1} \qquad \Longrightarrow \qquad q = 0.$$

Then

$$\|u\|_T := \left(|u|^p_{W^{k,p}(\Omega)} + \|Tu\|^p_Z\right)^{1/p}$$

is a norm equivalent to $\|u\|_{W^{k,p}(\Omega)}$.

Proof. Minkowski's inequality[3] gives the triangle inequality. If $\|u\|_T = 0$, the polynomial-kernel lemma gives $u \in P_{k-1}$, and $Tu = 0$ then implies $u = 0$. The upper bound $\|u\|_T \leq C\|u\|_{W^{k,p}(\Omega)}$ follows from the boundedness of T.

Suppose that the reverse bound fails. After normalization, there are $u_n \in W^{k,p}(\Omega)$ with $\|u_n\|_{W^{k,p}} = 1$ and $\|u_n\|_T \to 0$. By compactness of $W^{k,p}(\Omega) \hookrightarrow W^{k-1,p}(\Omega)$, a subsequence converges in $W^{k-1,p}(\Omega)$. Moreover,

$$\|u_n - u_m\|^p_{W^{k,p}} = \|u_n - u_m\|^p_{W^{k-1,p}} + |u_n - u_m|^p_{W^{k,p}} \to 0,$$

since the first term tends to zero by convergence in $W^{k-1,p}$ and $|u_n - u_m|_{W^{k,p}} \leq \|u_n\|_T + \|u_m\|_T \to 0$. Thus this subsequence converges in $W^{k,p}(\Omega)$ to some u with $\|u\|_{W^{k,p}} = 1$. But $|u|_{W^{k,p}} = 0$ and, by continuity of T, $Tu = 0$. The preceding lemma gives $u \in P_{k-1}$, contradicting the assumption on T. $\square$

For bounded linear functionals $\ell_1, \ldots, \ell_J$ on $W^{k,p}(\Omega)$, take $Z = \mathbb{R}^J$ with $\|(z_j)\|_Z = (\sum_{j=1}^J |z_j|^p)^{1/p}$ and $Tu = (\ell_1(u), \ldots, \ell_J(u))$. The theorem applies if no nonzero polynomial $q \in P_{k-1}$ satisfies $\ell_j(q) = 0$ for every $j = 1, \ldots, J$. Taking $Z = L^p(\Omega)$ and $Tu = u$ also gives

$$\|u\|_{W^{k,p}(\Omega)} \simeq \left(\|u\|^p_{L^p(\Omega)} + |u|^p_{W^{k,p}(\Omega)}\right)^{1/p}.$$

Here $\simeq$ means equivalence up to constants independent of u. The lower equivalence constant obtained by compactness is not explicit.

[3] $\|f + g\|_{L^p(\Omega)} \leq \|f\|_{L^p(\Omega)} + \|g\|_{L^p(\Omega)}, \qquad 1 \leq p < \infty$

Zero boundary values

Define $H_0^1(\Omega)$ as the closure of $C_0^\infty(\Omega)$ in $H^1(\Omega)$. It is a closed subspace of $H^1(\Omega)$.

Lemma 3.12 (Trace characterization of $H_0^1(\Omega)$)**.**
For a bounded Lipschitz domain,

$$H_0^1(\Omega) = \{u \in H^1(\Omega) : \gamma u = 0 \text{ on } \partial\Omega\}.$$

Every $u \in H_0^1(\Omega)$ has an extension by zero in $H^1(\mathbb{R}^d)$.

Proof (not examinable). For $\phi \in C_0^\infty(\Omega)$ we have $\gamma\phi = 0$. Continuity of the trace therefore gives $H_0^1(\Omega) \subset \ker\gamma$. Conversely, let $u \in H^1(\Omega)$ with $\gamma u = 0$, and extend u by zero to $\widetilde{u}$ on $\mathbb{R}^d$. The integration-by-parts formula for H^1 functions on a Lipschitz domain gives, for every $\varphi \in C_0^\infty(\mathbb{R}^d)$,

$$\int_{\mathbb{R}^d} \widetilde{u}\,\partial_i\varphi\,dx = -\int_\Omega (\partial_i u)\varphi\,dx + \int_{\partial\Omega} (\gamma u)\varphi n_i\,ds = -\int_{\mathbb{R}^d} \widetilde{\partial_i u}\,\varphi\,dx.$$

Here n_i is the i-th component of the outer unit normal and $\widetilde{\partial_i u}$ denotes extension by zero. Thus the weak derivatives of $\widetilde{u}$ are in $L^2(\mathbb{R}^d)$, and $\widetilde{u} \in H^1(\mathbb{R}^d)$.

Choose a finite partition of unity that splits $\widetilde{u}$ into one piece compactly supported inside Ω and pieces compactly supported in boundary charts. In each bi-Lipschitz chart, flatten the boundary, translate the zero-extended boundary piece a small distance into the half-space representing Ω, and map it back. The shifted piece is compactly supported inside Ω. Translations converge in H^1 and the bi-Lipschitz changes of variables preserve this convergence by the chain rule and change of variables. Mollifying each shifted piece with a radius smaller than its distance to $\partial\Omega$, and mollifying the interior piece directly, gives functions in $C_0^\infty(\Omega)$ converging to u in $H^1(\Omega)$. Hence $u \in H_0^1(\Omega)$. The zero-extension calculation also proves the final assertion. □

Remark 3.13. More generally, $H_0^k(\Omega)$ is the closure of $C_0^\infty(\Omega)$ in $H^k(\Omega)$. On a Lipschitz domain, it can be characterized by $\gamma(D^\alpha u) = 0$ for $|\alpha| \le k-1$. These traces are defined because $D^\alpha u \in H^1(\Omega)$.

Friedrichs and Poincaré inequalities

Lemma 3.14 (Friedrichs inequality)**.**
Let Ω be a bounded Lipschitz domain and $\Gamma_D \subset \partial\Omega$ measurable with positive

surface measure. Set

$$H^1_D(\Omega) := \{u \in H^1(\Omega) : (\gamma u)|_{\Gamma_D} = 0\}.$$

Then, with C_F depending on Ω and Γ_D,

$$\|u\|_{L^2(\Omega)} \le C_F \|\nabla u\|_{L^2(\Omega)} \qquad \forall u \in H^1_D(\Omega).$$

In particular, $\|\nabla \cdot\|_{L^2(\Omega)} = |\cdot|_{H^1(\Omega)}$ is an equivalent norm on $H^1_D(\Omega)$.

Proof. The trace theorem shows that $Tu := \int_{\Gamma_D} \gamma u \, ds$ is a bounded linear functional on $H^1(\Omega)$, since $|Tu| \le |\Gamma_D|^{1/2} C_{\mathrm{tr}} \|u\|_{H^1(\Omega)}$. For a constant q, $Tq = q|\Gamma_D|$, so T detects the kernel of the gradient seminorm. Applying the norm-equivalence theorem with $k = 1$ and $p = 2$ gives

$$\|u\|_{H^1(\Omega)} \le C\big(\|\nabla u\|^2_{L^2(\Omega)} + |Tu|^2\big)^{1/2}.$$

Since $Tu = 0$ on $H^1_D(\Omega)$, both assertions follow. □

Corollary 3.15 (Explicit bound for full Dirichlet conditions)**.**
For any bounded domain Ω, set $\operatorname{diam}(\Omega) := \sup_{x,y\in\Omega} |x - y|$. *Then*

$$\|u\|_{L^2(\Omega)} \le \operatorname{diam}(\Omega) \|\nabla u\|_{L^2(\Omega)} \qquad \forall u \in H^1_0(\Omega),$$

and consequently

$$\|\nabla u\|_{L^2(\Omega)} \le \|u\|_{H^1(\Omega)} \le \sqrt{1 + \operatorname{diam}(\Omega)^2}\, \|\nabla u\|_{L^2(\Omega)} \qquad \forall u \in H^1_0(\Omega).$$

Proof. Let $a = \inf_{x\in\Omega} x_1$, $b = \sup_{x\in\Omega} x_1$, and $L = b - a \le \operatorname{diam}(\Omega)$. For $u \in C^\infty_0(\Omega)$, extend u by zero to $\widetilde{u} \in C^\infty_0(\mathbb{R}^d)$. Writing $x = (x_1, x')$, the fundamental theorem of calculus and Cauchy–Schwarz give

$$|\widetilde{u}(x_1, x')|^2 = \left| \int_a^{x_1} \partial_1 \widetilde{u}(\xi, x') \, d\xi \right|^2 \le L \int_a^b |\partial_1 \widetilde{u}(\xi, x')|^2 \, d\xi.$$

Integration over $(a, b) \times \mathbb{R}^{d-1}$ yields $\|u\|_{L^2(\Omega)} \le L \|\partial_1 u\|_{L^2(\Omega)}$. Density gives the first estimate for $H^1_0(\Omega)$, and the H^1-norm identity gives the second. □

Remark 3.16 (Convex and star-shaped domains)**.**
A domain Ω is convex if $[x, \bar{x}] \subset \Omega$ for all $x, \bar{x} \in \Omega$. It is star-shaped with respect to a point $\bar{x} \in \Omega$ if $[x, \bar{x}] \subset \Omega$ for every $x \in \Omega$, and star-shaped with respect to a ball $B \subset \Omega$ if this holds for every $\bar{x} \in B$.

Lemma 3.17 (Classical Poincaré inequality)**.**
Let Ω be a bounded Lipschitz domain. For $u \in H^1(\Omega)$, set $\bar{u} := |\Omega|^{-1} \int_\Omega u\,dx$. Then

$$\|u - \bar{u}\|_{L^2(\Omega)} \le C_P \|\nabla u\|_{L^2(\Omega)} \qquad \forall u \in H^1(\Omega).$$

If Ω is star-shaped with respect to a ball $B \subset \Omega$, one may choose

$$C_P \le C\left(d, \frac{\operatorname{diam}(\Omega)}{\operatorname{diam}(B)}\right) \operatorname{diam}(\Omega).$$

If Ω is convex, $C_P \le \operatorname{diam}(\Omega)/\pi$.

Proof. The functional $Tu = \int_\Omega u\,dx$ is bounded on $H^1(\Omega)$ and is nonzero on every nonzero constant. Apply the norm-equivalence theorem with $k = 1$, $p = 2$ to $u - \bar{u}$, whose mean is zero. This gives $\|u - \bar{u}\|_{L^2} \le \|u - \bar{u}\|_{H^1} \le C\|\nabla u\|_{L^2}$. The geometric bounds require additional arguments: see Brenner and Scott, *The Mathematical Theory of Finite Element Methods*, Lemma 4.3.14, for star-shaped domains, and Bebendorf, *A Note on the Poincaré Inequality for Convex Domains* (2003), Theorem 3.2, for the convex bound. □

Remark 3.18 (Other measurements of constants)**.**
Let Ω be a bounded Lipschitz domain and $1 \le p < \infty$. For any measurable $V \subset \Omega$ of positive volume or measurable $S \subset \partial\Omega$ of positive surface measure, set either $Tu = \int_V u\,dx$ or $Tu = \int_S \gamma u\,ds$. Hölder's inequality (and the trace theorem for S) shows that T is bounded on $W^{1,p}(\Omega)$, while $Tq \neq 0$ for every nonzero constant q. The norm-equivalence theorem therefore gives

$$\|u\|_{W^{1,p}(\Omega)} \simeq \left(|u|^p_{W^{1,p}(\Omega)} + |Tu|^p\right)^{1/p}.$$

Remark 3.19 (Poincaré and polynomial approximation)**.**
The Poincaré inequality approximates $u \in H^1(\Omega)$ by the constant $\bar{u}$ with an error controlled by ∇u. Approximation by higher-degree polynomials is the subject of the Bramble–Hilbert lemma, used below for interpolation error estimates.

4 Weak solutions to elliptic boundary value problems

We now demonstrate the application of the theoretical results presented in the last two chapters to the analysis of the weak formulation of elliptic boundary value problems for partial differential equations presented in the first chapter.

4.1 Model problem

We start by considering our model problem from Chapter 1, which is

$$-\operatorname{div}(a\nabla u) = f \qquad \text{in } \Omega, \tag{4.1}$$
$$u = 0 \qquad \text{on } \partial\Omega. \tag{4.2}$$

For the following analysis, we will assume that $\Omega \subset \mathbb{R}^d$ is a bounded Lipschitz domain and that $f \in L^2(\Omega)$ and $a \in L^\infty(\Omega)$ with

$$0 < \underline{a} \le a(x) \le \overline{a} \qquad \text{for almost every } x \in \Omega. \tag{4.3}$$

Step 1. Let us recall the derivation of the weak formulation. For this formal calculation, first assume that a, u, and the test function v are sufficiently smooth and that $v = 0$ on $\partial\Omega$. Multiply (4.1) by v, integrate over Ω, and use integration by parts. This gives

$$\begin{aligned}\int_\Omega f\, v\, dx &= \int_\Omega -\operatorname{div}(a\nabla u)\, v\, dx \\ &= \int_\Omega a\nabla u \cdot \nabla v\, dx - \int_{\partial\Omega} \vec{n} \cdot (a\nabla u)\, v\, ds,\end{aligned}$$

with $\vec{n}$ denoting the outward unit normal vector on $\partial\Omega$. Since Ω is a Lipschitz domain, $\vec{n}$ is well-defined almost everywhere, and the boundary term vanishes because $v = 0$ there. The identity then extends to test functions in $H_0^1(\Omega)$ by density. The remaining volume integral makes sense for $u \in H^1(\Omega)$, and from (4.2) we see that we should look for $u \in H_0^1(\Omega)$. This leads to

Definition 4.1. The variational problem

$$\text{Find } u \in H_0^1(\Omega): \quad \int_\Omega a\nabla u \cdot \nabla v \, dx = \int_\Omega f\, v \, dx \quad \forall v \in H_0^1(\Omega) \qquad \text{(VP)}$$

is called the *weak formulation*, and u is called a *weak solution* of (4.1)–(4.2).

Step 2. The weak formulation is an abstract variational problem, as considered in Chapter 2, with $V = H_0^1(\Omega)$, $\mathcal{A}(u,v) := \int_\Omega a\nabla u \cdot \nabla v \, dx$ and $\ell(v) := \int_\Omega f\, v \, dx$. To show existence of a unique weak solution, we thus only have to verify the conditions of the Lax-Milgram theorem.

(i) By construction, the space $V = H_0^1(\Omega) = \overline{C_0^\infty(\Omega)}^{\|\cdot\|_{H^1(\Omega)}}$ is a Hilbert space.

(ii) The mapping $\mathcal{A} : V \times V \to \mathbb{R}$ is bilinear and

$$|\mathcal{A}(u,v)| \le \|a\|_{L^\infty(\Omega)} \|\nabla u\|_{L^2(\Omega)} \|\nabla v\|_{L^2(\Omega)} \le \overline{a}\, \|u\|_{H^1(\Omega)} \|v\|_{H^1(\Omega)},$$

which even holds for all $u, v \in H^1(\Omega)$. This shows that $\mathcal{A}$ is continuous with continuity constant $\overline{a}$. Moreover, we have

$$\mathcal{A}(u,u) = \int_\Omega a\,|\nabla u|^2 \, dx \ge \underline{a} \|\nabla u\|_{L^2(\Omega)}^2 \ge \tfrac{\underline{a}}{1+C_F^2} \|u\|_{H^1(\Omega)}^2 \qquad \forall u \in H_0^1(\Omega).$$

In the last step, we used $\|u\|_{L^2(\Omega)} \le C_F \|\nabla u\|_{L^2(\Omega)}$ from the Friedrichs inequality and thus made use of the boundary values (4.2). Hence $\mathcal{A}$ is elliptic with constant $\alpha = \underline{a}/(1 + C_F^2)$.

(iii) The mapping $\ell : V \to \mathbb{R}$, $\ell(v) = \int_\Omega f\, v \, dx$ is linear and

$$|\ell(v)| = \left| \int_\Omega f\, v \, dx \right| \le \|f\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} \le \|f\|_{L^2(\Omega)} \|v\|_{H^1(\Omega)}$$

for all $v \in H^1(\Omega)$. This shows that ℓ is indeed a bounded linear functional on the space $V = H_0^1(\Omega)$ with $\|\ell\|_{V^*} \le \|f\|_{L^2(\Omega)}$.

Step 3. The Lax–Milgram lemma therefore gives a unique solution $u \in H_0^1(\Omega)$ of (VP), with $\|u\|_{H^1(\Omega)} \le \frac{1+C_F^2}{\underline{a}} \|f\|_{L^2(\Omega)}$. This is also the unique weak solution of (4.1)–(4.2).

Remarks on higher regularity

For $\Omega = (0,1)$, the model problem reduces to

$$\begin{aligned} -\partial_x(a\partial_x u) &= f \qquad \text{in } (0,1), \\ u(0) = u(1) &= 0. \end{aligned}$$

The weak equation gives $(au')' = -f$ in the distributional sense. Since $au' \in L^2(0,1)$, this implies $au' \in H^1(0,1)$. The assumptions $a \in W^{1,\infty}(0,1)$ and $a \geq \underline{a} > 0$ imply $1/a \in W^{1,\infty}(0,1)$, so $u' = (au')/a \in H^1(0,1)$. In particular, $u \in H^2(0,1)$ and

$$\partial_{xx} u = -\tfrac{1}{a}(f + \partial_x a\, \partial_x u) \qquad \text{almost everywhere.}$$

Moreover,

$$\|u''\|_{L^2(0,1)} \leq \tfrac{1}{\underline{a}}\big(\|f\|_{L^2(0,1)} + \|a'\|_{L^\infty(0,1)}\|u'\|_{L^2(0,1)}\big) \leq C\|f\|_{L^2(0,1)},$$

where the last step uses the preceding H^1 estimate. A similar statement also holds in multiple dimensions. Some additional conditions on the regularity of the domain are, however, required.

Lemma 4.2 (H^2- and H^k-regularity)**.**

(i) Let $\Omega \subset \mathbb{R}^d$ be a bounded convex domain and $a \in W^{1,\infty}(\Omega)$. Then the weak solution of (4.1)–(4.2) *satisfies $\|u\|_{H^2(\Omega)} \leq C\|f\|_{L^2(\Omega)}$.*

(ii) Let $k \geq 2$ be an integer. If Ω is a $C^{k-1,1}$-domain (its local boundary charts have Lipschitz continuous derivatives of order $k-1$) and $a \in W^{k-1,\infty}(\Omega)$, then for any $f \in H^{k-2}(\Omega)$, we have $u \in H^k(\Omega)$ and $\|u\|_{H^k(\Omega)} \leq C\|f\|_{H^{k-2}(\Omega)}$.

For details, we refer to the books of Evans: *Partial differential equations* or Grisvard: *Elliptic problems in non-smooth domains.*

Remark 4.3. (i) Under appropriate smoothness assumptions on the coefficient a and the domain Ω, we conclude that $f \in H^{k-2}(\Omega)$ implies $u \in H^k(\Omega)$. By Sobolev's embedding theorem, we obtain *classical solutions* for k large.

(ii) On the L-shaped domain $\Omega = (-1,1)^2 \setminus [0,1]^2$, the solution of $-\Delta u = 1$ with $u = 0$ on $\partial\Omega$ does not lie in $H^2(\Omega)$. Reentrant corners (or edges in 3D) cause *singularities* in the solution.

(iii) Similar statements also hold for other types of boundary conditions, but note that changing boundary conditions may also cause singularities.

For $0 < s < 1$, the fractional Sobolev space $H^s(\Omega)$ consists of functions $w \in L^2(\Omega)$ with finite norm

$$\|w\|^2_{H^s(\Omega)} := \|w\|^2_{L^2(\Omega)} + \int_\Omega \int_\Omega \frac{|w(x) - w(y)|^2}{|x-y|^{d+2s}}\, dx\, dy.$$

We write $H^{1+s}(\Omega)$ for the functions in $H^1(\Omega)$ whose first derivatives belong to $H^s(\Omega)$, with norm $\|u\|^2_{H^{1+s}(\Omega)} := \|u\|^2_{H^1(\Omega)} + \sum_{i=1}^d \|\partial_i u\|^2_{H^s(\Omega)}$. On a domain with

reentrant corners, a solution may lie in $H^{1+s}(\Omega)$ for some $s < 1$ while failing to lie in $H^2(\Omega)$. This intermediate regularity enters finite element error estimates later on.

4.2 Trace spaces and generalized data

Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain and let $\Gamma \subset \partial\Omega$ be a measurable boundary part. Further, let $\gamma_\Gamma u := (\gamma u)|_\Gamma$, where $\gamma : H^1(\Omega) \to L^2(\partial\Omega)$ is the trace operator from Chapter 3. The traces on Γ form the space

$$H^{1/2}(\Gamma) := \gamma_\Gamma(H^1(\Omega)), \qquad \|g\|_{H^{1/2}(\Gamma)} := \inf_{\gamma_\Gamma w = g} \|w\|_{H^1(\Omega)}.$$

Lemma 4.4 (Trace space and minimum-norm lifting)**.**
The space $H^{1/2}(\Gamma)$ equipped with the quotient norm is a Hilbert space. Every $g \in H^{1/2}(\Gamma)$ has a unique minimum-norm lifting $u_g \in (\ker \gamma_\Gamma)^\perp$ with

$$\gamma_\Gamma u_g = g, \qquad \|u_g\|_{H^1(\Omega)} = \|g\|_{H^{1/2}(\Gamma)}.$$

The map $g \mapsto u_g$ is linear and isometric. Moreover, the trace satisfies

$$\|\gamma_\Gamma u\|_{H^{1/2}(\Gamma)} \le \|u\|_{H^1(\Omega)} \qquad \forall u \in H^1(\Omega). \tag{4.4}$$

Proof. Set $K := \ker \gamma_\Gamma$. The restricted trace $\gamma_\Gamma : H^1(\Omega) \to L^2(\Gamma)$ is continuous, so K is a closed subspace of $H^1(\Omega)$. The projection lemma from Chapter 2 therefore gives $H^1(\Omega) = K \oplus K^\perp$. For a given $g \in H^{1/2}(\Gamma)$, choose any lifting $w \in H^1(\Omega)$ with $\gamma_\Gamma w = g$. Decompose it as

$$w = w_0 + w_\perp, \qquad w_0 \in K, \quad w_\perp \in K^\perp.$$

Since $\gamma_\Gamma w_0 = 0$, the orthogonal component $w_\perp$ still has trace g. Every other lifting $\widetilde{w}$ of g has the form $\widetilde{w} = w_\perp + v$ with $v \in K$, because their difference has zero trace on Γ. Pythagoras gives

$$\|\widetilde{w}\|^2_{H^1(\Omega)} = \|w_\perp\|^2_{H^1(\Omega)} + \|v\|^2_{H^1(\Omega)} \ge \|w_\perp\|^2_{H^1(\Omega)}.$$

Thus $u_g := w_\perp$ is the unique minimum-norm lifting of g. Consequently, $\gamma_\Gamma : K^\perp \to H^{1/2}(\Gamma)$ is a linear isometric bijection. The inner product

$$(g, h)_{H^{1/2}(\Gamma)} := (u_g, u_h)_{H^1(\Omega)}$$

induces precisely the quotient norm defined above. Since $K^\perp$ is a closed subspace of the Hilbert space $H^1(\Omega)$, it is complete. The isometry therefore makes $H^{1/2}(\Gamma)$ a Hilbert space as well.

Finally, for any $u \in H^1(\Omega)$, the function u itself is an admissible lifting of its trace $\gamma_\Gamma u$. Taking the infimum over all such liftings proves (4.4). □

For $\Gamma = \emptyset$, we use the convention $H^{1/2}(\Gamma) = \{0\}$. For $d \geq 2$, the full trace space $H^{1/2}(\partial\Omega)$ is in general a proper subspace of $L^2(\partial\Omega)$: arbitrary L^2 boundary data need not possess an H^1 lifting.

The natural space for weak boundary loads on Γ is its dual,

$$H^{-1/2}(\Gamma) := H^{1/2}(\Gamma)^*, \qquad \|h\|_{H^{-1/2}(\Gamma)} := \sup_{0 \neq w \in H^{1/2}(\Gamma)} \frac{|\langle h, w \rangle_\Gamma|}{\|w\|_{H^{1/2}(\Gamma)}}.$$

Every $h \in L^2(\Gamma)$ defines such a functional through

$$\langle h, w \rangle_\Gamma := \int_\Gamma hw \, ds.$$

Indeed, for any lifting u of w, the trace theorem gives

$$\|w\|_{L^2(\Gamma)} = \|\gamma_\Gamma u\|_{L^2(\Gamma)} \leq C_{\mathrm{tr}} \|u\|_{H^1(\Omega)}.$$

Taking the infimum over all liftings gives $\|w\|_{L^2(\Gamma)} \leq C_{\mathrm{tr}} \|w\|_{H^{1/2}(\Gamma)}$, so $L^2(\Gamma)$ embeds continuously into $H^{-1/2}(\Gamma)$. The opposite does not hold in general.

4.2.1 Dirichlet data and minimum-norm lifting

The quotient norm measures the smallest H^1 norm of a lifting. For full boundary data, the minimum-norm lifting is characterized by

$$-\Delta u + u = 0 \qquad \text{in } \Omega, \tag{4.5}$$

$$u = g \qquad \text{on } \partial\Omega. \tag{4.6}$$

A weak solution satisfies $\gamma u = g$ and

$$\int_\Omega (\nabla u \cdot \nabla v + uv) \, dx = 0 \qquad \forall v \in H_0^1(\Omega).$$

The left-hand side is precisely $(u, v)_{H^1(\Omega)}$. Since $\ker \gamma = H_0^1(\Omega)$, the weak equation says that u is orthogonal to the kernel of the trace in the full H^1 inner product.

Lemma 4.5. *Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain. For every $g \in H^{1/2}(\partial\Omega)$ there is a unique weak solution of (4.5)–(4.6). It is the unique minimum-norm lifting of g, and*

$$\|u\|_{H^1(\Omega)} = \|g\|_{H^{1/2}(\partial\Omega)}.$$

Proof. Apply Lemma 4.4 with $\Gamma = \partial\Omega$. The minimum-norm lifting u_g belongs to $(\ker\gamma)^\perp$. Since $\ker\gamma = H_0^1(\Omega)$, it satisfies the weak equation. Conversely, every weak solution is a lifting in $(\ker\gamma)^\perp$ and hence equals u_g. The norm identity follows from the minimum-norm property proved above. □

4.2.2 Generalized volume loads

For a full Dirichlet problem, define

$$H^{-1}(\Omega) := H_0^1(\Omega)^*, \qquad \|\ell\|_{H^{-1}(\Omega)} := \sup_{0\neq v\in H_0^1(\Omega)} \frac{|\ell(v)|}{\|v\|_{H^1(\Omega)}}.$$

Every $\ell \in H^{-1}(\Omega)$ defines a distribution by restriction to $C_0^\infty(\Omega)$. Friedrichs and Lax–Milgram show that

$$\text{Find } u \in H_0^1(\Omega): \qquad \int_\Omega \nabla u \cdot \nabla v \, dx = \ell(v) \qquad \forall v \in H_0^1(\Omega)$$

has a unique solution with $\|u\|_{H^1(\Omega)} \le C_\Omega \|\ell\|_{H^{-1}(\Omega)}$.

For example, $\vec{F} \in L^2(\Omega)^d$ defines

$$\ell_{\vec{F}}(v) := -\int_\Omega \vec{F} \cdot \nabla v \, dx, \qquad \|\ell_{\vec{F}}\|_{H^{-1}(\Omega)} \le \|\vec{F}\|_{L^2(\Omega)}.$$

In distributions, $\ell_{\vec{F}} = \operatorname{div}\vec{F}$, since both act on $\varphi \in C_0^\infty(\Omega)$ as $-\int_\Omega \vec{F} \cdot \nabla\varphi \, dx$. Thus

$$-\Delta u = \operatorname{div}\vec{F} \qquad \text{in } \Omega, \tag{4.7}$$

$$u = 0 \qquad \text{on } \partial\Omega \tag{4.8}$$

has a unique weak solution even when $\operatorname{div}\vec{F}$ is not an L^2 function. The space $H^{-1}(\Omega)$ is tied to the full Dirichlet test space $H_0^1(\Omega)$. For a mixed problem with test space V_0, the natural total load space is instead V_0^*.

4.3 Diffusion, convection, and reaction with mixed boundary conditions

Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain and split its boundary, up to sets of surface measure zero, into disjoint measurable parts Γ_D, Γ_N, and Γ_R. We consider

$$-\operatorname{div}(a\nabla u) + \vec{b}\cdot\nabla u + cu = f \qquad \text{in } \Omega, \tag{4.9}$$

$$u = g \qquad \text{on } \Gamma_D, \tag{4.10}$$

$$\vec{n}\cdot(a\nabla u) = h_N \qquad \text{on } \Gamma_N, \tag{4.11}$$

$$\vec{n}\cdot(a\nabla u) + \eta u = h_R \qquad \text{on } \Gamma_R. \tag{4.12}$$

For a boundary part Γ, the notation $|\Gamma|$ denotes its surface measure. Here $a, c \in L^\infty(\Omega)$, $\vec{b} \in L^\infty(\Omega)^d$, $\eta \in L^\infty(\Gamma_R)$, and

$$f \in L^2(\Omega), \qquad g \in H^{1/2}(\Gamma_D), \qquad h_N \in H^{-1/2}(\Gamma_N), \qquad h_R \in H^{-1/2}(\Gamma_R).$$

The term $\vec{b}\cdot\nabla u$ is in nonconservative form, so the corresponding classical normal flux is $\vec{n}\cdot(a\nabla u)$. For nonsmooth solutions, the Neumann and Robin conditions in (4.11)–(4.12) are understood through the weak formulation below rather than as pointwise traces of $a\nabla u$. Write $\gamma_D = \gamma_{\Gamma_D}$ and set

$$V_0 := \{v \in H^1(\Omega) : \gamma_D v = 0\}, \qquad V_g := \{u \in H^1(\Omega) : \gamma_D u = g\}.$$

If $\Gamma_D = \emptyset$, we set $g = 0$ and $V_0 = V_g = H^1(\Omega)$. For $u, v \in H^1(\Omega)$ define

$$\mathcal{A}(u,v) := \int_\Omega \big(a\nabla u\cdot\nabla v + (\vec{b}\cdot\nabla u)v + cuv\big)\,dx + \int_{\Gamma_R} \eta\,(\gamma_{\Gamma_R}u)(\gamma_{\Gamma_R}v)\,ds.$$

For the data above, define the total load on V_0 by

$$\ell(v) := \int_\Omega fv\,dx + \langle h_N, \gamma_{\Gamma_N}v\rangle_{\Gamma_N} + \langle h_R, \gamma_{\Gamma_R}v\rangle_{\Gamma_R}.$$

Cauchy–Schwarz, the duality estimate, and (4.4) show that $\ell \in V_0^*$, with

$$\|\ell\|_{V_0^*} \le \|f\|_{L^2(\Omega)} + \|h_N\|_{H^{-1/2}(\Gamma_N)} + \|h_R\|_{H^{-1/2}(\Gamma_R)}.$$

For smooth data and solutions, integration by parts gives

$$\int_\Omega -\operatorname{div}(a\nabla u)v\,dx = \int_\Omega a\nabla u\cdot\nabla v\,dx - \int_{\partial\Omega} \vec{n}\cdot(a\nabla u)\,v\,ds.$$

Choosing $v \in V_0$ and substituting the boundary conditions gives $\mathcal{A}(u,v) = \ell(v)$ with this total load. More generally, we allow any bounded linear functional $\ell \in V_0^*$ and consider the weak formulation

$$\text{Find } u \in V_g\text{:} \qquad \mathcal{A}(u,v) = \ell(v) \qquad \forall v \in V_0. \tag{VP$_g$}$$

For the concrete load above, this identity defines the weak meaning of the natural Neumann and Robin conditions. The Dirichlet condition is imposed through the trial space and is called *essential*.

Homogenization. Choose the minimum-norm lifting $u_g \in V_g$. Then

$$\|u_g\|_{H^1(\Omega)} = \|g\|_{H^{1/2}(\Gamma_D)}.$$

For $\Gamma_D = \emptyset$, take $u_g = 0$. Since $V_g = u_g + V_0$, writing $u = u_g + u_0$ turns (VP_g) into

$$\text{Find } u_0 \in V_0\text{:} \qquad \mathcal{A}(u_0, v) = \ell(v) - \mathcal{A}(u_g, v) =: \ell_0(v) \qquad \forall v \in V_0. \qquad (\mathrm{VP}_0)$$

Lemma 4.6 (Well-posedness with mixed boundary conditions)**.**
Suppose that

$$\begin{aligned} a \geq \underline{a} > 0, \qquad c, \eta \geq 0 \quad &\textit{almost everywhere,} \\ |\Gamma_D| > 0 \quad \textit{or} \quad \int_\Omega c\,dx + \int_{\Gamma_R} \eta\,ds > 0. \qquad &(\mathrm{A}) \end{aligned}$$

Then there is a constant $\alpha_0 > 0$, independent of $\vec{b}$, such that if $\|\vec{b}\|_{L^\infty(\Omega)} < \alpha_0$, problem (VP_g) has a unique solution for all $\ell \in V_0^*$ and $g \in H^{1/2}(\Gamma_D)$, and

$$\|u\|_{H^1(\Omega)} \leq C\big(\|\ell\|_{V_0^*} + \|g\|_{H^{1/2}(\Gamma_D)}\big).$$

The constant depends on the domain, the boundary parts, the coefficient bounds, and the positive coercivity margin $\alpha_0 - \|\vec{b}\|_{L^\infty(\Omega)}$.

Proof. The trace theorem makes $V_0 = \ker \gamma_D$ a closed subspace of $H^1(\Omega)$, hence a Hilbert space. For $u, v \in H^1(\Omega)$, Cauchy–Schwarz and the trace theorem give

$$\begin{aligned} |\mathcal{A}(u, v)| &\leq C_{\mathcal{A}} \|u\|_{H^1(\Omega)} \|v\|_{H^1(\Omega)}, \\ C_{\mathcal{A}} &:= \|a\|_{L^\infty(\Omega)} + \|\vec{b}\|_{L^\infty(\Omega)} + \|c\|_{L^\infty(\Omega)} + C_{\mathrm{tr}}^2 \|\eta\|_{L^\infty(\Gamma_R)}. \end{aligned}$$

The load ℓ is continuous on V_0 by assumption. For $v \in V_0$, set

$$E(v) := \int_\Omega \big(a|\nabla v|^2 + cv^2\big)\,dx + \int_{\Gamma_R} \eta\,(\gamma_{\Gamma_R} v)^2\,ds.$$

We claim that $E(v) \geq \alpha_0 \|v\|_{H^1(\Omega)}^2$ for some $\alpha_0 > 0$. If $|\Gamma_D| > 0$, the Friedrichs inequality from Chapter 3 gives $\|v\|_{H^1}^2 \leq (1 + C_F^2)\|\nabla v\|_{L^2}^2$, so we may take $\alpha_0 = \underline{a}/(1 + C_F^2)$. Otherwise, the second alternative in (A) applies. The bounded map

$$T : H^1(\Omega) \longrightarrow L^2(\Omega) \times L^2(\Gamma_R), \qquad Tv := \big(\sqrt{c}\,v, \sqrt{\eta}\,\gamma_{\Gamma_R} v\big)$$

has the product norm $\|Tv\|^2 = \|\sqrt{c}\,v\|^2_{L^2(\Omega)} + \|\sqrt{\eta}\,\gamma_{\Gamma_R} v\|^2_{L^2(\Gamma_R)}$. It is nonzero on every nonzero constant. The Sobolev norm-equivalence theorem from Chapter 3 therefore gives $\|v\|^2_{H^1} \le C(\|\nabla v\|^2_{L^2} + \|Tv\|^2)$. Since $a \ge \underline{a}$, this proves the claim also in this case.

The convection term satisfies

$$\begin{aligned}\left|\int_\Omega (\vec{b}\cdot\nabla v)v\,dx\right| &\le \|\vec{b}\|_{L^\infty(\Omega)}\|\nabla v\|_{L^2(\Omega)}\|v\|_{L^2(\Omega)}\\ &\le \|\vec{b}\|_{L^\infty(\Omega)}\|v\|^2_{H^1(\Omega)}.\end{aligned}$$

Consequently $\mathcal{A}(v,v) \ge \alpha\|v\|^2_{H^1}$ on V_0, where $\alpha := \alpha_0 - \|\vec{b}\|_{L^\infty(\Omega)} > 0$. Moreover,

$$\|\ell_0\|_{V_0^*} \le \|\ell\|_{V_0^*} + C_{\mathcal{A}}\|u_g\|_{H^1(\Omega)}.$$

Lax–Milgram applied to (VP_0) gives a unique u_0 with $\|u_0\|_{H^1} \le \alpha^{-1}\|\ell_0\|_{V_0^*}$. The triangle inequality, the choice of u_g, and the preceding bounds give the stated estimate for $u = u_g + u_0$.

If two liftings are chosen, their resulting solutions have the same Dirichlet trace. Their difference $w \in V_0$ satisfies $\mathcal{A}(w,v) = 0$ for every $v \in V_0$. Testing with $v = w$ and using coercivity gives $w = 0$, so the solution is independent of the lifting. □

Remark 4.7. The theorem includes square-integrable Neumann and Robin data through the continuous embedding

$$L^2(\Gamma) \hookrightarrow H^{-1/2}(\Gamma).$$

If $\Gamma_D = \emptyset$, $c = 0$, and $\vec{b} = 0$, a nonnegative Robin coefficient with $\int_{\Gamma_R}\eta\,ds > 0$ still yields a unique solution. If $\eta = 0$ as well, constants remain in the kernel (see the pure Neumann problem below).

For full Dirichlet conditions, another alternative to the smallness assumption on $\vec{b}$ is $\vec{b} \in W^{1,\infty}(\Omega)^d$ and $\operatorname{div}\vec{b} \le 0$. Integration by parts gives

$$\int_\Omega (\vec{b}\cdot\nabla v)v\,dx = -\tfrac{1}{2}\int_\Omega (\operatorname{div}\vec{b})v^2\,dx \ge 0 \qquad \forall v \in H^1_0(\Omega).$$

4.4 The pure Neumann problem

Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain. For $f \in L^2(\Omega)$ and $h \in H^{-1/2}(\partial\Omega)$, consider, in the weak sense,

$$-\Delta u = f \qquad \text{in } \Omega, \tag{4.13}$$

$$\partial_n u = h \qquad \text{on } \partial\Omega. \tag{4.14}$$

These data define a total load on $H^1(\Omega)$ by

$$\ell(v) := \int_\Omega f v \, dx + \langle h, \gamma v \rangle_{\partial\Omega}.$$

Cauchy–Schwarz, the duality estimate, and (4.4) give

$$\|\ell\|_{H^1(\Omega)^*} \le \|f\|_{L^2(\Omega)} + \|h\|_{H^{-1/2}(\partial\Omega)}.$$

Here $\partial_n u = h$ denotes the natural boundary condition encoded by the variational equation

$$\int_\Omega \nabla u \cdot \nabla v \, dx = \ell(v) \qquad \forall v \in H^1(\Omega).$$

More generally, we allow any bounded linear functional $\ell \in H^1(\Omega)^*$. Taking $v = 1$ gives the necessary compatibility condition

$$\ell(1) = 0. \tag{4.15}$$

For the concrete load above, this means

$$\int_\Omega f \, dx + \langle h, 1 \rangle_{\partial\Omega} = 0,$$

where 1 in the boundary pairing denotes the trace of the constant function with value one. Solutions are determined only up to an additive constant. To select one, set

$$V_\diamond := \left\{ v \in H^1(\Omega) : \int_\Omega v \, dx = 0 \right\},$$

and solve

$$\text{Find } u \in V_\diamond: \qquad \int_\Omega \nabla u \cdot \nabla v \, dx = \ell(v) \qquad \forall v \in V_\diamond. \tag{VP$_n$}$$

Lemma 4.8. *For every $\ell \in H^1(\Omega)^*$, problem* (VP$_n$) *has a unique solution, with*

$$\|u\|_{H^1(\Omega)} \le C_\Omega \|\ell\|_{H^1(\Omega)^*}.$$

If (4.15) *holds, this function satisfies the weak equation for every $v \in H^1(\Omega)$. In that case, every other weak solution is of the form $u + c$ with $c \in \mathbb{R}$. For the concrete load above, these are the weak solutions of* (4.13)–(4.14).

Proof. The mean is a bounded functional on $H^1(\Omega)$, so $V_\diamond$ is a closed Hilbert subspace. The Poincaré inequality from Chapter 3 gives, for $v \in V_\diamond$,

$$\|v\|^2_{H^1(\Omega)} \leq (1 + C_P^2)\|\nabla v\|^2_{L^2(\Omega)}.$$

Moreover, the load satisfies

$$|\ell(v)| \leq \|\ell\|_{H^1(\Omega)^*}\|v\|_{H^1(\Omega)} \qquad \forall v \in V_\diamond.$$

Lax–Milgram therefore gives the unique solution and its estimate.

If the compatibility condition holds, write any $v \in H^1(\Omega)$ as $v = (v - \bar{v}) + \bar{v}$, where $\bar{v} = |\Omega|^{-1} \int_\Omega v\, dx$. The gradients of v and $v - \bar{v}$ agree, and

$$\ell(v) = \ell(v - \bar{v}) + \bar{v}\ell(1) = \ell(v - \bar{v}).$$

Thus the equation for $v - \bar{v} \in V_\diamond$ also gives the weak equation for v. Finally, the difference of two solutions has zero gradient and is therefore constant because Ω is connected. □

With similar reasoning, one can treat higher-order problems or systems of equations, such as the biharmonic problem and linear elasticity; see the exercises.

5 The Finite Element Method

We now discretize the model problem from Chapters 1 and 4,

$$-\operatorname{div}(a\nabla u) = f \qquad \text{in } \Omega, \tag{5.1}$$

$$u = 0 \qquad \text{on } \partial\Omega. \tag{5.2}$$

Throughout this chapter, $\Omega \subset \mathbb{R}^2$ is a bounded polygonal Lipschitz domain. We assume that $f \in L^2(\Omega)$ and $a \in L^\infty(\Omega)$ with $0 < \underline{a} \le a(x) \le \overline{a}$ almost everywhere. The polygonal boundary allows us to partition Ω exactly into straight-sided triangles.

With $V = H_0^1(\Omega)$, the weak formulation reads

$$\text{Find } u \in V\text{:} \qquad \mathcal{A}(u,v) = \ell(v) \qquad \forall v \in V, \tag{5.3}$$

where

$$\mathcal{A}(u,v) = \int_\Omega a\nabla u \cdot \nabla v \, dx, \qquad \ell(v) = \int_\Omega f v \, dx.$$

Chapter 4 provides continuity and ellipticity constants $C_A, \alpha > 0$ for $\mathcal{A}$ on V. For any finite-dimensional subspace $V_h \subset V$, the Galerkin approximation

$$\text{Find } u_h \in V_h\text{:} \qquad \mathcal{A}(u_h,v_h) = \ell(v_h) \qquad \forall v_h \in V_h \tag{5.4}$$

therefore has a unique solution. By Céa's lemma from Chapter 2,

$$\|u - u_h\|_{H^1(\Omega)} \le \frac{C_A}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|_{H^1(\Omega)}. \tag{5.5}$$

Our task is thus to construct spaces V_h with good approximation properties.

Outline. We first construct continuous piecewise linear functions on triangular meshes and their nodal interpolant. Bramble–Hilbert and scaling then give interpolation estimates, which yield finite element error bounds and convergence. Finally, we use duality to estimate the error in $L^2(\Omega)$ and discuss further estimates. The construction and analysis are presented for P_1 elements in two dimensions. Concrete implementation is the subject of the next chapter.

5.1 Triangulations and conforming finite element spaces

Definition 5.1 (Triangulation, shape regularity, and quasiuniformity)**.**
A *triangulation* $\mathcal{T}_h$ of Ω is a finite set of closed, non-degenerate triangles with

$$\overline{\Omega} = \bigcup_{T \in \mathcal{T}_h} T, \qquad \mathring{T} \cap \mathring{T}' = \emptyset \quad \text{for } T \neq T'.$$

It is called *conforming* if the intersection of two distinct triangles is either empty, a common vertex, or a common edge.

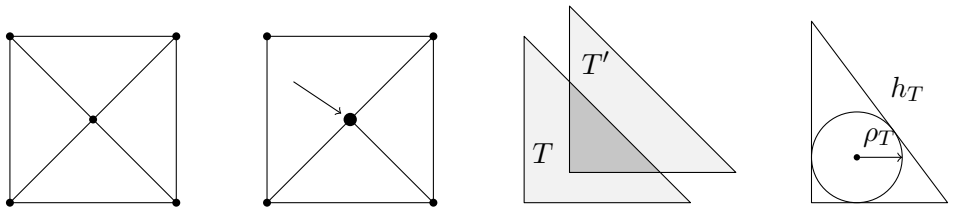
For $T \in \mathcal{T}_h$, let $h_T = \operatorname{diam}(T)$ and let ρ_T be its inradius. We set $h = \max_{T \in \mathcal{T}_h} h_T$. The triangulation is called γ*-shape regular* if

$$\rho_T \geq \gamma h_T \qquad \forall T \in \mathcal{T}_h$$

for some $\gamma > 0$. It is called *quasiuniform* if

$$\delta h \leq h_T \leq h \qquad \forall T \in \mathcal{T}_h$$

for some $\delta > 0$. For a family of meshes, shape regularity and quasiuniformity require the respective constants γ and δ to be independent of the mesh.



(a) Conforming triangulation (b) Hanging node (c) Overlapping triangles (d) Diameter and inradius

Figure 5.1: Mesh geometry: (a) a conforming triangulation; (b) a triangulation that is not conforming; (c) triangles with overlapping interiors, which cannot belong to a triangulation; (d) the diameter h_T is the longest edge length, and ρ_T is the inradius.

We consider conforming, γ-shape regular triangulations throughout. Conformity excludes hanging nodes. Shape regularity prevents arbitrarily small angles but allows elements of different sizes, whereas quasiuniformity additionally requires comparable sizes. A shape-regular family requires a common positive lower bound for ρ_T / h_T.

Example 5.2 (Uniform refinement)**.**
Start from a conforming, γ-shape regular triangulation $\mathcal{T}_0$. To obtain $\mathcal{T}_{\ell+1}$ from $\mathcal{T}_\ell$, connect the three edge midpoints in every triangle. Each triangle is thereby divided into four similar triangles. Since all elements are refined, the subdivisions agree on shared edges, so every $\mathcal{T}_\ell$ is conforming. Similarity preserves the ratio ρ_T/h_T, and hence the same shape-regularity constant works for every level. With $h_\ell = \max_{T\in\mathcal{T}_\ell} h_T$, we have

$$h_\ell = 2^{-\ell} h_0 \to 0 \qquad \text{as } \ell \to \infty.$$

Thus uniform refinement produces a concrete mesh family satisfying the assumptions of the convergence results below.

Let $\mathcal{N}_h$ denote the set of mesh vertices, and set

$$\mathcal{N}_h^0 = \mathcal{N}_h \cap \Omega, \qquad \mathcal{N}_h^\partial = \mathcal{N}_h \cap \partial\Omega.$$

These are the interior and boundary vertices, respectively.

Definition 5.3 (Piecewise polynomial spaces)**.**
For an integer $k \geq 0$, let $P_k = \operatorname{span}\{x^i y^j : i, j \geq 0,\ i + j \leq k\}$ and define

$$P_k(\mathcal{T}_h) = \{v \in L^2(\Omega) : v|_{\mathring{T}} \in P_k \text{ for every } T \in \mathcal{T}_h\},$$
$$S_k^0(\mathcal{T}_h) = P_k(\mathcal{T}_h) \cap C(\overline{\Omega}).$$

The conforming P_1 space for (5.3) is

$$V_h = S_1^0(\mathcal{T}_h) \cap H_0^1(\Omega). \tag{5.6}$$

The gluing result from Chapter 3 shows that $S_k^0(\mathcal{T}_h) \subset H^1(\Omega)$: the element restrictions are smooth, and their traces agree on common edges. To describe these functions explicitly, we use their nodal values.

Lemma 5.4 (Nodal basis)**.**
For every $x_i \in \mathcal{N}_h$, there exists a unique function $\phi_i \in S_1^0(\mathcal{T}_h)$ with

$$\phi_i(x_j) = \delta_{ij} \qquad \forall x_j \in \mathcal{N}_h.$$

These functions form a basis of $S_1^0(\mathcal{T}_h)$, and

$$v_h = \sum_{x_i \in \mathcal{N}_h} v_h(x_i)\phi_i \qquad \forall v_h \in S_1^0(\mathcal{T}_h). \tag{5.7}$$

Moreover, the functions ϕ_i associated with interior vertices form a basis of V_h.

Proof. An affine function on a non-degenerate triangle is uniquely determined by its values at the three vertices. Prescribing the nodal values therefore defines ϕ_i on each triangle. On a common edge, the two restrictions are affine functions with the same endpoint values, so they agree on the whole edge. Thus ϕ_i is globally continuous.

The same argument proves (5.7). Evaluating at the vertices shows that the functions ϕ_i are linearly independent. Finally, a function in $S_1^0(\mathcal{T}_h)$ vanishes on $\partial\Omega$ if and only if all its boundary nodal values vanish: its restriction to each boundary edge is affine. The trace characterization of $H_0^1(\Omega)$ from Chapter 3 then proves the last assertion. □

The functions ϕ_i are called *hat functions*. The support of ϕ_i is the union of the triangles containing x_i. Consequently, basis functions with disjoint supports do not interact in the Galerkin matrix, which leads to a sparse matrix.

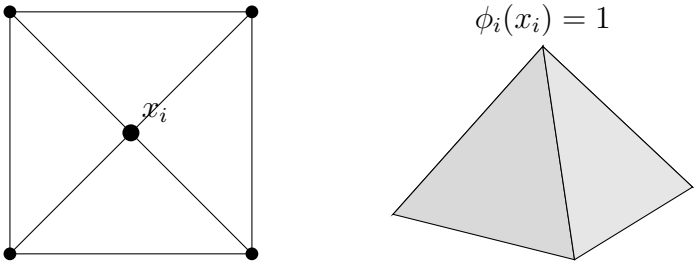


Figure 5.2: A conforming triangulation and the hat function at its interior vertex.

5.2 Nodal interpolation and polynomial approximation

For a continuous function u, we define its *nodal interpolant* by

$$I_h : C(\overline{\Omega}) \to S_1^0(\mathcal{T}_h), \qquad I_h u = \sum_{x_i \in \mathcal{N}_h} u(x_i)\phi_i. \tag{5.8}$$

Thus $(I_h u)(x_i) = u(x_i)$ at every vertex. The restriction $(I_h u)|_T$ is the local affine interpolant $I_T(u|_T)$, determined by the three vertex values on T. In particular,

$$I_T q = q \qquad \forall q \in P_1(T).$$

If u is continuous and vanishes on $\partial\Omega$, then $I_h u \in V_h$.

Point values are not defined for general functions in $H^1(\Omega)$ in two dimensions. However, the Sobolev embedding from Chapter 3 gives $H^2(\Omega) \hookrightarrow C(\overline{\Omega})$. Fractional regularity is discussed in Remark 5.15. For $u \in H^2(\Omega) \cap H_0^1(\Omega)$, we have $I_h u \in V_h$, and therefore

$$\inf_{v_h \in V_h} \|u - v_h\|_{H^1(\Omega)} \leq \|u - I_h u\|_{H^1(\Omega)}.$$

Combined with Céa's lemma, this bounds the Galerkin error by the interpolation error in H^1. Our aim is to show that the interpolation error is of order h in H^1 and of order h^2 in L^2. Transferring the latter rate to the Galerkin error will require additional regularity of the dual problem. We first prove an estimate on a fixed reference triangle and then identify the powers of h by scaling.

The Bramble–Hilbert lemma

The Poincaré inequality bounds the error of approximation by constants using first derivatives. The following result extends this principle to higher-degree polynomials.

Lemma 5.5 (Bramble–Hilbert)**.**
Let $\omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain, $k \geq 1$ an integer, $1 \leq p < \infty$, and Z a normed space. Suppose that $L : W^{k,p}(\omega) \to Z$ is linear and satisfies

$$\|Lv\|_Z \leq C_L \|v\|_{W^{k,p}(\omega)} \quad \text{and} \quad Lq = 0 \qquad \forall q \in P_{k-1}(\omega).$$

Then

$$\|Lv\|_Z \leq C_{\omega,k,p} C_L |v|_{W^{k,p}(\omega)} \qquad \forall v \in W^{k,p}(\omega).$$

Here $P_{k-1}(\omega)$ consists of the polynomials of total degree at most $k-1$, restricted to ω.

Proof. For $|\alpha| \leq k-1$, define the bounded linear functionals

$$\ell_\alpha(v) = \int_\omega D^\alpha v \, dx.$$

Their boundedness follows from Hölder's inequality. They determine a polynomial $q \in P_{k-1}(\omega)$ uniquely: if all its moments $\ell_\alpha(q)$ vanish, derivatives of order $k-1$ first show that the coefficients of degree $k-1$ vanish. Repeating this argument with decreasing degree gives $q = 0$. Since the number of moments equals $\dim P_{k-1}$, there is a unique polynomial q with

$$\ell_\alpha(q) = \ell_\alpha(v) \qquad \forall |\alpha| \leq k-1.$$

By the norm-equivalence theorem, Lemma 3.11, the expression

$$\left(|w|^p_{W^{k,p}(\omega)} + \sum_{|\alpha|\le k-1} |\ell_\alpha(w)|^p \right)^{1/p}$$

is equivalent to $\|w\|_{W^{k,p}(\omega)}$. Applied to $w = v - q$, this yields

$$\|v-q\|_{W^{k,p}(\omega)} \le C_{\omega,k,p}|v-q|_{W^{k,p}(\omega)} = C_{\omega,k,p}|v|_{W^{k,p}(\omega)}.$$

Finally, $Lq = 0$ and boundedness of L give

$$\|Lv\|_Z = \|L(v-q)\|_Z \le C_L\|v-q\|_{W^{k,p}(\omega)},$$

which proves the assertion. □

For example, $Lv = v - \bar{v}$ with $\bar{v} = |\omega|^{-1}\int_\omega v\,dx$ recovers Poincaré for $k = 1$ and $p = 2$. For interpolation, we instead apply the lemma to $L = \mathrm{id} - I$, provided that I is bounded and reproduces the required polynomials.

Interpolation on the reference triangle

Let

$$\widehat{T} = \mathrm{conv}\{(0,0),(1,0),(0,1)\}.$$

Its nodal basis is

$$\widehat{\phi}_1(\xi) = 1 - \xi_1 - \xi_2, \qquad \widehat{\phi}_2(\xi) = \xi_1, \qquad \widehat{\phi}_3(\xi) = \xi_2.$$

These are the barycentric coordinates of $\xi \in \widehat{T}$: they are nonnegative and sum to one. Denote the corresponding nodal interpolation operator by $\widehat{I}$.

Lemma 5.6 (Reference interpolation estimate)**.**
There exists a constant $\widehat{C}$ depending only on $\widehat{T}$ such that

$$\|\widehat{u} - \widehat{I}\widehat{u}\|_{H^1(\widehat{T})} \le \widehat{C}|\widehat{u}|_{H^2(\widehat{T})} \qquad \forall \widehat{u} \in H^2(\widehat{T}).$$

Proof. The embedding $H^2(\widehat{T}) \hookrightarrow C(\overline{\widehat{T}})$ makes the vertex evaluations bounded. Hence

$$\|\widehat{I}\widehat{u}\|_{H^1(\widehat{T})} \le \sum_{i=1}^{3} |\widehat{u}(\widehat{x}_i)|\|\widehat{\phi}_i\|_{H^1(\widehat{T})} \le C\|\widehat{u}\|_{H^2(\widehat{T})}.$$

Thus $L = \mathrm{id} - \widehat{I} : H^2(\widehat{T}) \to H^1(\widehat{T})$ is bounded. Since $\widehat{I}q = q$ for all $q \in P_1$, the assertion follows from Lemma 5.5 with $k = p = 2$. □

5.3 Scaling and interpolation error estimates

The reference estimate contains no mesh size. To obtain an estimate on a triangle T, we transform to $\widehat{T}$, apply Lemma 5.6, and transform back. Figure 5.3 shows the geometric correspondence.

Lemma 5.7 (Affine transformation and scaling)**.**
Let T have vertices x_1, x_2, x_3. The affine map

$$F_T(\xi) = x_1 + B_T\xi, \qquad B_T = (x_2 - x_1,\ x_3 - x_1),$$

maps $\widehat{T}$ bijectively onto T. For a γ-shape regular triangle,

$$\|B_T\|_F \le Ch_T, \qquad \|B_T^{-1}\|_F \le C_\gamma h_T^{-1}, \qquad c_\gamma h_T^2 \le |\det B_T| \le Ch_T^2. \tag{5.9}$$

Here $\|\cdot\|_F$ is the Frobenius matrix norm. If $\widehat{u} = u \circ F_T$ with $u \in H^2(T)$, then

$$\|\widehat{u}\|_{L^2(\widehat{T})} = |\det B_T|^{-1/2}\|u\|_{L^2(T)}, \tag{5.10}$$

$$|u|_{H^1(T)} \le |\det B_T|^{1/2}\|B_T^{-1}\|_F|\widehat{u}|_{H^1(\widehat{T})}, \tag{5.11}$$

$$|\widehat{u}|_{H^2(\widehat{T})} \le C|\det B_T|^{-1/2}\|B_T\|_F^2|u|_{H^2(T)}. \tag{5.12}$$

Proof. Both columns of B_T have length at most h_T, so $\|B_T\|_F \le \sqrt{2}h_T$. Moreover, $|\det B_T| = 2|T|$. The triangle contains a disk of radius $\rho_T \ge \gamma h_T$, and $|T| \le h_T^2/2$. These facts give the determinant bounds. In two dimensions, the inverse matrix formula yields

$$\|B_T^{-1}\|_F = \frac{\|B_T\|_F}{|\det B_T|},$$

which proves the remaining bound in (5.9). The change-of-variables formula gives (5.10). For smooth u, the chain rule gives

$$(\nabla u) \circ F_T = B_T^{-\top}\widehat{\nabla}\widehat{u},$$
$$\widehat{\nabla}^2\widehat{u} = B_T^\top((\nabla^2 u) \circ F_T)B_T.$$

Here $\nabla^2 u$ is the Hessian. Its L^2 norm with the pointwise Frobenius norm is equivalent to $|u|_{H^2}$. Estimating the matrix products and changing variables proves (5.11)–(5.12). Density of smooth functions in $H^2(T)$ extends these formulas and estimates to every $u \in H^2(T)$. □

Interpolation commutes with this affine transformation:

$$(I_T u) \circ F_T = \widehat{I}(u \circ F_T). \tag{5.13}$$

Indeed, both sides are affine on $\widehat{T}$ and have the same values at its three vertices.

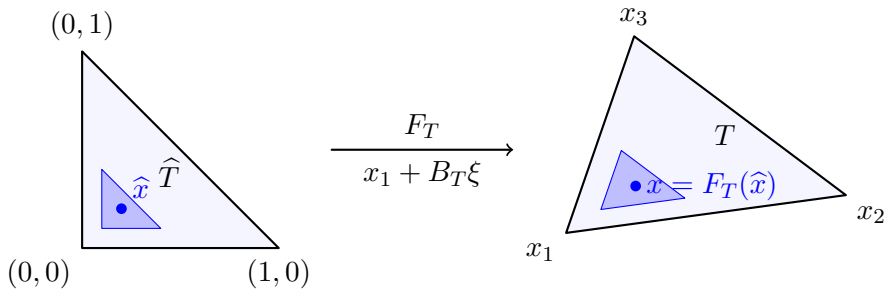


Figure 5.3: Affine reference-to-physical mapping with corresponding point and subtriangle. Areas scale by $|\det B_T|$ and gradients by $B_T^{-\top}$.

Lemma 5.8 (Local interpolation error estimates)**.**
For every γ-shape regular triangle T and $u \in H^2(T)$,

$$\|u - I_T u\|_{L^2(T)} \leq C h_T^2 |u|_{H^2(T)}, \tag{5.14}$$
$$|u - I_T u|_{H^1(T)} \leq C_\gamma h_T |u|_{H^2(T)}. \tag{5.15}$$

The constants are independent of h_T and u, and the first estimate does not require a bound on the shape regularity.

Proof. Set $\widehat{u} = u \circ F_T$ and $\widehat{e} = \widehat{u} - \widehat{I}\widehat{u}$. By (5.13), $\widehat{e} = (u - I_T u) \circ F_T$. The reference estimate and scaling give

$$\begin{aligned}
\|u - I_T u\|_{L^2(T)} &= |\det B_T|^{1/2} \|\widehat{e}\|_{L^2(\widehat{T})} \\
&\leq C |\det B_T|^{1/2} |\widehat{u}|_{H^2(\widehat{T})} \leq C \|B_T\|_F^2 |u|_{H^2(T)}, \\
|u - I_T u|_{H^1(T)} &\leq |\det B_T|^{1/2} \|B_T^{-1}\|_F |\widehat{e}|_{H^1(\widehat{T})} \\
&\leq C \|B_T^{-1}\|_F \|B_T\|_F^2 |u|_{H^2(T)}.
\end{aligned}$$

The bounds for B_T and B_T^{-1} yield the assertions. □

The scaling argument in three steps. The preceding proof follows a pattern that also applies to inverse and trace estimates:

1. Transform the left-hand side from T to $\widehat{T}$, keeping track of the Jacobian factor and the factors B_T^{-1} arising from derivatives.
2. Apply an estimate on the fixed reference element. Here, Bramble–Hilbert bounds the interpolation error by the H^2 seminorm.
3. Transform the right-hand side back to T and use the geometric bounds for B_T and B_T^{-1} to identify the powers of h_T.

Here the determinant factors cancel. The remaining factor $\|B_T\|_F^2$ gives h_T^2 in L^2, and the additional factor $\|B_T^{-1}\|_F$ gives h_T in the H^1 seminorm.

The use of a seminorm in (5.15) makes the scaling explicit. For the full norm, the two estimates give

$$\|u - I_T u\|_{H^1(T)} \leq C_\gamma h_T (1 + h_T^2)^{1/2} |u|_{H^2(T)}.$$

Since $h_T \leq \operatorname{diam}(\Omega)$, this factor is uniformly bounded.

Lemma 5.9 (Global interpolation error estimates)**.**
For $u \in H^2(\Omega)$ and shape regular triangulation $\mathcal{T}_h$, the nodal interpolant satisfies

$$\|u - I_h u\|_{L^2(\Omega)}^2 \leq C \sum_{T \in \mathcal{T}_h} h_T^4 |u|_{H^2(T)}^2, \tag{5.16}$$

$$|u - I_h u|_{H^1(\Omega)}^2 \leq C_\gamma \sum_{T \in \mathcal{T}_h} h_T^2 |u|_{H^2(T)}^2. \tag{5.17}$$

In particular,

$$\|u - I_h u\|_{L^2(\Omega)} \leq C h^2 |u|_{H^2(\Omega)} \qquad \text{and} \qquad \|u - I_h u\|_{H^1(\Omega)} \leq C_{\gamma,\Omega} h |u|_{H^2(\Omega)}.$$

Proof. The global interpolant restricts to $I_T u$ on each triangle. Since $u - I_h u \in H^1(\Omega)$, the squared L^2 norm and H^1 seminorm equal the sums of their element contributions. Summing Lemma 5.8 gives (5.16)–(5.17). The final estimates follow from $h_T \leq h \leq \operatorname{diam}(\Omega)$. □

Remark 5.10 (Piecewise regularity)**.**
Define the broken Sobolev space and its seminorm by

$$H^2(\mathcal{T}_h) = \{v \in L^2(\Omega) : v|_{\mathring{T}} \in H^2(\mathring{T}) \text{ for every } T \in \mathcal{T}_h\},$$
$$|v|_{H^2(\mathcal{T}_h)}^2 = \sum_{T \in \mathcal{T}_h} |v|_{H^2(T)}^2.$$

The proof of Lemma 5.9 only uses $u \in H^1(\Omega) \cap H^2(\mathcal{T}_h)$. To see that nodal interpolation is still well-defined, take the continuous representative on each triangle. Global H^1 regularity forces the traces to agree almost everywhere on common edges. The continuous traces therefore agree everywhere on each edge, including its endpoints. Since the triangles incident to a vertex form an edge-connected fan in a conforming triangulation of a Lipschitz polygon, their vertex values also agree.

Thus u has a continuous representative on $\overline{\Omega}$, and the preceding estimates remain valid with the broken seminorm. If also $u \in H_0^1(\Omega)$, then $I_h u \in V_h$.

5.4 Error estimates and convergence in H^1

We can now estimate the finite element error by combining the approximation results with Céa's lemma, (5.5). This only requires continuity and ellipticity of $\mathcal{A}$. Symmetry is not needed for the following H^1 error estimate or the convergence argument.

Lemma 5.11 (Error estimate in H^1)**.**
Let $u \in V$ solve (5.3), and assume that $u \in H^2(\mathcal{T}_h)$. Its finite element approximation $u_h \in V_h$ satisfies

$$\|u - u_h\|_{H^1(\Omega)} \leq C \left(\sum_{T \in \mathcal{T}_h} h_T^2 |u|_{H^2(T)}^2 \right)^{1/2} \leq Ch|u|_{H^2(\mathcal{T}_h)}. \tag{5.18}$$

The constant depends on Ω, γ, and the continuity and ellipticity bounds, but is independent of h and u.

Proof. Remark 5.10 gives $I_h u \in V_h$. Céa's lemma and Friedrichs' inequality for $u - I_h u \in H_0^1(\Omega)$ yield

$$\begin{aligned} \|u - u_h\|_{H^1(\Omega)} &\leq \frac{C_A}{\alpha} \|u - I_h u\|_{H^1(\Omega)} \leq C |u - I_h u|_{H^1(\Omega)} \\ &\leq C \left(\sum_{T \in \mathcal{T}_h} h_T^2 |u|_{H^2(T)}^2 \right)^{1/2}. \end{aligned}$$

Using $h_T \leq h$ proves the second bound. □

Remark 5.12 (Energy norm in the symmetric case)**.**
For the scalar diffusion problem considered in this chapter, $\mathcal{A}$ is symmetric and therefore defines an inner product on V. The associated *energy norm*

$$\|v\|_{\mathcal{A}} = \sqrt{\mathcal{A}(v, v)}$$

is equivalent to the H^1 norm on V. Galerkin orthogonality then gives the best-approximation identity

$$\|u - u_h\|_{\mathcal{A}} = \inf_{v_h \in V_h} \|u - v_h\|_{\mathcal{A}} \leq \|u - I_h u\|_{\mathcal{A}},$$

whenever $I_h u \in V_h$. Thus Lemma 5.11 also gives an error estimate in the energy norm for this model problem.

Remark 5.13 (Regularity and local refinement)**.**
If Ω is convex and $a \in W^{1,\infty}(\Omega)$, Chapter 4 gives the regularity bound

$$\|u\|_{H^2(\Omega)} \leq C\|f\|_{L^2(\Omega)}.$$

Combining this with Lemma 5.11 yields

$$\|u - u_h\|_{H^1(\Omega)} \leq Ch\|f\|_{L^2(\Omega)}.$$

The local sum in (5.18) suggests using smaller elements where second derivatives are large. This motivates local mesh refinement.

Lemma 5.14 (Convergence without additional regularity)**.**
Let $\{\mathcal{T}_h\}$ be a family of conforming triangulations with a common shape-regularity constant and $h \to 0$. Then, for every $u \in H_0^1(\Omega)$,

$$\inf_{v_h \in V_h} \|u - v_h\|_{H^1(\Omega)} \to 0.$$

Thus, the Galerkin solutions of (5.4) *converge to the weak solution in $H^1(\Omega)$.*

Proof. Fix $\varepsilon > 0$. By the definition of $H_0^1(\Omega)$, there exists $w \in C_0^\infty(\Omega)$ with $\|u - w\|_{H^1(\Omega)} < \varepsilon/2$. Since w vanishes on the boundary, $I_h w \in V_h$, and

$$\begin{aligned}\inf_{v_h \in V_h} \|u - v_h\|_{H^1(\Omega)} &\leq \|u - w\|_{H^1(\Omega)} + \|w - I_h w\|_{H^1(\Omega)} \\ &< \varepsilon/2 + Ch|w|_{H^2(\Omega)}.\end{aligned}$$

The second term is less than $\varepsilon/2$ for every sufficiently small h. This proves the approximation statement, and Céa's lemma then gives convergence of the Galerkin solutions. □

Remark 5.15 (Convergence with fractional regularity)**.**
The fractional Sobolev spaces from Chapter 4 describe intermediate regularity. Let $u \in H^{1+s}(\Omega) \cap V$, $0 < s \leq 1$, be the weak solution. On a family of conforming meshes with a common shape-regularity constant, one has

$$\|u - u_h\|_{H^1(\Omega)} \leq C_s h^s \|u\|_{H^{1+s}(\Omega)}.$$

The constant C_s is independent of h and u, but may depend on s, Ω, the shape-regularity constant, and the continuity and ellipticity bounds. For $s = 1$, this follows from the preceding H^1 error estimate. For $0 < s < 1$, it follows from an interpolation estimate in fractional Sobolev spaces and Céa's lemma. In two

dimensions, $H^{1+s}(\Omega) \hookrightarrow C(\overline{\Omega})$ for every $s > 0$, so nodal interpolation is still well-defined. We omit the fractional interpolation argument. For an alternative approach by operator interpolation, see Brenner and Scott, *The Mathematical Theory of Finite Element Methods*, 3rd ed. (2008), Section 14.3, Theorem 14.3.3 and the subsequent discussion of Dirichlet boundary conditions (pp. 376–377). Additional regularity beyond H^2 does not in general improve the order h in the H^1 norm for P_1 elements.

5.5 Duality and error estimates in weaker norms

The variational formulation on $V = H_0^1(\Omega)$ makes the H^1 norm the natural norm for error analysis. The interpolation estimate in L^2, however, suggests a faster rate in that weaker norm. To transfer this improvement to the Galerkin solution, we use a dual problem.

Lemma 5.16 (Aubin–Nitsche duality argument)**.**
Assume that for every $g \in L^2(\Omega)$ the solution $z \in V$ of

$$\mathcal{A}(v, z) = (g, v)_{L^2(\Omega)} \qquad \forall v \in V \tag{5.19}$$

belongs to $H^2(\Omega)$ and satisfies

$$\|z\|_{H^2(\Omega)} \le C_{\mathrm{reg}} \|g\|_{L^2(\Omega)}. \tag{5.20}$$

Then the Galerkin error satisfies

$$\|u - u_h\|_{L^2(\Omega)} \le Ch \|u - u_h\|_{H^1(\Omega)}. \tag{5.21}$$

If in addition $u \in H^2(\mathcal{T}_h)$, then

$$\|u - u_h\|_{L^2(\Omega)} \le Ch^2 |u|_{H^2(\mathcal{T}_h)}. \tag{5.22}$$

The constants are independent of h and may depend on C_{reg}.

Proof. Set $e = u - u_h$ and solve (5.19) with $g = e$. Existence and uniqueness follow from Lax–Milgram. Since $z \in H^2(\Omega) \cap V$, its interpolant satisfies $I_h z \in V_h$. Galerkin orthogonality therefore gives

$$\begin{aligned} \|e\|_{L^2(\Omega)}^2 &= \mathcal{A}(e, z) = \mathcal{A}(e, z - I_h z) \\ &\le C_A \|e\|_{H^1(\Omega)} \|z - I_h z\|_{H^1(\Omega)} \\ &\le Ch \|e\|_{H^1(\Omega)} |z|_{H^2(\Omega)} \le C C_{\mathrm{reg}} h \|e\|_{H^1(\Omega)} \|e\|_{L^2(\Omega)}. \end{aligned}$$

If $e = 0$, the claim is immediate. Otherwise, division by $\|e\|_{L^2(\Omega)}$ proves (5.21). Using Lemma 5.11 gives (5.22). □

Remark 5.17 (The role of dual regularity)**.**
For our symmetric model problem, (5.19) is the weak formulation of

$$\begin{aligned} -\operatorname{div}(a\nabla z) &= g \qquad \text{in } \Omega, \\ z &= 0 \qquad \text{on } \partial\Omega. \end{aligned}$$

Thus (5.20) follows from Chapter 4 when Ω is convex and $a \in W^{1,\infty}(\Omega)$.

The proof separates two sources of regularity: approximation of the primal solution controls the H^1 error, and approximation of the dual solution yields the extra factor h. For a nonsymmetric coercive bilinear form, the dual solution must still occur in the second argument, as in (5.19).

Remark 5.18 (Quantities of interest)**.**
Suppose that we are interested in a functional $G \in V^*$, e.g., the average

$$G(v) = \frac{1}{|D|}\int_D v\,dx$$

over a measurable subset $D \subset \Omega$ of positive area. Define $z_G \in V$ by

$$\mathcal{A}(v, z_G) = G(v) \qquad \forall v \in V.$$

For every $z_h \in V_h$, Galerkin orthogonality gives

$$G(u) - G(u_h) = \mathcal{A}(u - u_h, z_G - z_h).$$

Consequently,

$$|G(u) - G(u_h)| \le C_A \|u - u_h\|_{H^1(\Omega)} \inf_{z_h \in V_h} \|z_G - z_h\|_{H^1(\Omega)}.$$

If $u, z_G \in H^2(\mathcal{T}_h)$, the preceding approximation estimates yield

$$|G(u) - G(u_h)| \le Ch^2 |u|_{H^2(\mathcal{T}_h)} |z_G|_{H^2(\mathcal{T}_h)}.$$

Thus the accuracy depends on the approximability of its associated dual solution.

5.6 Further estimates and extensions

The preceding sections provide the basic convergence theory for conforming P_1 elements. We conclude with extensions of the approximation argument and an application of inverse inequalities.

Higher polynomial degree and reduced regularity

For a fixed degree $k \geq 1$, consider $V_h^k = S_k^0(\mathcal{T}_h) \cap H_0^1(\Omega)$, using the usual Lagrange nodes on each triangle. A polynomial in P_k is determined by its values at the barycentric nodes with coordinates $(i/k, j/k, \ell/k)$, where i, j, ℓ are nonnegative integers with $i + j + \ell = k$. For $k = 2$, these are the vertices and edge midpoints.

For an integer $2 \leq m \leq k + 1$, the reference interpolation operator is bounded on $H^m(\widehat{T})$ and reproduces P_{m-1}. Applying Bramble–Hilbert and scaling as above gives

$$\|u - I_h^k u\|_{L^2(\Omega)} \leq C h^m |u|_{H^m(\Omega)},$$
$$\|u - I_h^k u\|_{H^1(\Omega)} \leq C h^{m-1} |u|_{H^m(\Omega)}.$$

For $u \in H^m(\Omega) \cap V$, Céa's lemma therefore yields the same h^{m-1} rate for the Galerkin error in H^1. More generally, for integer regularity $m \geq 2$, the rate is $h^{\min\{k, m-1\}}$. The constants depend additionally on k and m. Details of the higher-order interpolation are left to the exercises.

Inverse inequalities and a maximum-norm estimate

On a fixed finite-dimensional polynomial space, a stronger norm can be bounded by a weaker one. Scaling these bounds to a physical element introduces negative powers of its diameter.

Lemma 5.19 (Inverse inequalities)**.**
For $v_h \in P_1(T)$ on a γ-shape regular triangle,

$$\|v_h\|_{L^\infty(T)} \leq C_\gamma h_T^{-1} \|v_h\|_{L^2(T)},$$
$$|v_h|_{H^1(T)} \leq C_\gamma h_T^{-1} \|v_h\|_{L^2(T)}.$$

Proof. Set $\widehat{v}_h = v_h \circ F_T$. Norm equivalence on $P_1(\widehat{T})$ gives

$$\|\widehat{v}_h\|_{L^\infty(\widehat{T})} + |\widehat{v}_h|_{H^1(\widehat{T})} \leq C \|\widehat{v}_h\|_{L^2(\widehat{T})}.$$

For the first estimate, use

$$\|v_h\|_{L^\infty(T)} = \|\widehat{v}_h\|_{L^\infty(\widehat{T})} \leq C \|\widehat{v}_h\|_{L^2(\widehat{T})} \leq C |\det B_T|^{-1/2} \|v_h\|_{L^2(T)} \leq C_\gamma h_T^{-1} \|v_h\|_{L^2(T)}.$$

For the second, (5.11) and (5.10) instead yield

$$|v_h|_{H^1(T)} \leq C \|B_T^{-1}\|_F \|v_h\|_{L^2(T)} \leq C_\gamma h_T^{-1} \|v_h\|_{L^2(T)}.$$

□

These inequalities hold for discrete functions, not for arbitrary functions in the corresponding Sobolev spaces! The same argument applies to $P_k(T)$, with constants that additionally depend on k.

Lemma 5.20 (Maximum-norm interpolation estimates)**.**
For a γ-shape regular triangle T,

$$\|u - I_T u\|_{L^\infty(T)} \le C_\gamma h_T |u|_{H^2(T)} \qquad \forall u \in H^2(T),$$
$$\|u - I_T u\|_{L^\infty(T)} \le C h_T^2 |u|_{W^{2,\infty}(T)} \qquad \forall u \in W^{2,\infty}(T).$$

Proof. The embedding $H^2(\widehat{T}) \hookrightarrow C(\overline{\widehat{T}})$ and the nodal representation show that $\mathrm{id} - \widehat{I} : H^2(\widehat{T}) \to L^\infty(\widehat{T})$ is bounded and annihilates P_1. Bramble–Hilbert with $k = p = 2$ and $Z = L^\infty(\widehat{T})$, followed by scaling, gives

$$\|u - I_T u\|_{L^\infty(T)} \le C |\widehat{u}|_{H^2(\widehat{T})} \le C |\det B_T|^{-1/2} \|B_T\|_F^2 |u|_{H^2(T)} \le C_\gamma h_T |u|_{H^2(T)}.$$

For the second estimate, take the $C^{1,1}$ representative of $u \in W^{2,\infty}(T)$. The mean value theorem applied to ∇u on the convex triangle T gives

$$|\nabla u(y) - \nabla u(x)| \le C |u|_{W^{2,\infty}(T)} |y - x| \qquad \forall x, y \in T.$$

Fix $x \in T$ and set $d_i = x_i - x$ for its three vertices x_i. The fundamental theorem of calculus along the segment from x to x_i yields the Taylor formula

$$u(x_i) = u(x) + \nabla u(x) \cdot d_i + R_i, \qquad R_i = \int_0^1 \big(\nabla u(x + t d_i) - \nabla u(x)\big) \cdot d_i \, dt.$$

The gradient bound and $|d_i| \le h_T$ imply

$$|R_i| \le C |u|_{W^{2,\infty}(T)} |d_i|^2 \int_0^1 t \, dt \le C h_T^2 |u|_{W^{2,\infty}(T)}.$$

The barycentric coordinates satisfy $\lambda_i(x) \ge 0$, $\sum_i \lambda_i(x) = 1$, and $\sum_i \lambda_i(x) d_i = 0$.[1] Multiplying the Taylor formulas by $\lambda_i(x)$ and summing therefore cancels the linear terms and gives

$$|I_T u(x) - u(x)| = \left| \sum_{i=1}^3 \lambda_i(x) R_i \right| \le \sum_{i=1}^3 \lambda_i(x) |R_i| \le C h_T^2 |u|_{W^{2,\infty}(T)}.$$

Taking the supremum over $x \in T$ proves the second estimate. This argument does not require shape regularity. □

[1] By definition, $x = \sum_i \lambda_i(x) x_i$ and $\sum_i \lambda_i(x) = 1$. Hence $\sum_i \lambda_i(x) d_i = \sum_i \lambda_i(x) x_i - x \sum_i \lambda_i(x) = x - x = 0$.

To pass from a local inverse estimate to a global one with a single mesh size, we now additionally assume quasiuniformity as in Definition 5.1. Since $h_T \geq \delta h$, Lemma 5.19 implies

$$\|v_h\|_{L^\infty(\Omega)} \leq C_{\gamma,\delta} h^{-1} \|v_h\|_{L^2(\Omega)} \qquad \forall v_h \in S_1^0(\mathcal{T}_h).$$

Lemma 5.21 (An elementary L^∞ error estimate)**.**
Let the assumptions of Lemma 5.11 hold. Assume in addition the dual regularity estimate (5.20) *and quasiuniformity of the mesh. Then*

$$\|u - u_h\|_{L^\infty(\Omega)} \leq Ch|u|_{H^2(\mathcal{T}_h)},$$

where the constant may also depend on δ and C_{reg}.

Proof. Split the error into an interpolation error and a discrete function:

$$\|u - u_h\|_{L^\infty(\Omega)} \leq \|u - I_h u\|_{L^\infty(\Omega)} + \|I_h u - u_h\|_{L^\infty(\Omega)}.$$

The first term is bounded by $Ch|u|_{H^2(\mathcal{T}_h)}$ using Lemma 5.20 on each triangle. For the second, the global inverse inequality gives

$$\begin{aligned} \|I_h u - u_h\|_{L^\infty(\Omega)} &\leq Ch^{-1} \|I_h u - u_h\|_{L^2(\Omega)} \\ &\leq Ch^{-1} \big(\|I_h u - u\|_{L^2(\Omega)} + \|u - u_h\|_{L^2(\Omega)} \big) \\ &\leq Ch|u|_{H^2(\mathcal{T}_h)}. \end{aligned}$$

Here the last step uses the interpolation estimate and Aubin–Nitsche. Combining the bounds proves the assertion. □

Remark 5.22 (A sharper maximum-norm estimate)**.**
The elementary estimate above is not sharp for smoother solutions. For example, for the Poisson problem ($a = 1$) on a convex polygon Ω, a family of conforming, shape-regular and quasiuniform meshes, and $u \in W^{2,\infty}(\Omega) \cap V$, one has

$$\|u - u_h\|_{L^\infty(\Omega)} \leq Ch^2 \big(1 + |\log h|\big) \|u\|_{W^{2,\infty}(\Omega)} \qquad (0 < h \leq 1).$$

The constant is independent of h and u. Compared with the order h^2 interpolation estimate, the Galerkin error bound contains an additional logarithmic factor. Its proof requires more refined analysis than the inverse-inequality argument above. See Brenner and Scott, *The Mathematical Theory of Finite Element Methods*, 3rd ed. (2008), Section 8.5 (p. 230), combined with Corollary 8.1.12 (p. 217).

Remark 5.23 (Other boundary conditions)**.**
For homogeneous Dirichlet conditions on only part of the boundary, assume that this part is a union of mesh edges. One then restricts the nodal basis at all vertices of those edges. For a coercive Robin problem from Chapter 4, one uses the full space $S_1^0(\mathcal{T}_h)$. The local interpolation arguments are unchanged, and Céa's lemma again gives an H^1 error estimate. Error estimates in weaker norms require regularity of the dual problem with the corresponding boundary conditions.

Summary of the error estimates

Table 5.1 summarizes the Galerkin error estimates for the P_1 method. All rows assume $u \in V = H_0^1(\Omega)$ and conforming meshes with a common shape-regularity constant. The mesh size is $h = \max_{T \in \mathcal{T}_h} h_T$. The rates indicate the powers of h in the estimates. The corresponding norms or seminorms of u and constants independent of h are suppressed.

Error norm	Additional regularity of u	Dual regularity	Rate	Quasiuniformity required
$H^1(\Omega)$	$H^2(\mathcal{T}_h)$	—	h	no
$H^1(\Omega)$	$H^{1+r}(\Omega)$	—	h^r	no
$L^2(\Omega)$	$H^2(\mathcal{T}_h)$	H^2	h^2	no
$L^2(\Omega)$	$H^{1+r}(\Omega)$	H^{1+s}	h^{r+s}	no
$L^\infty(\Omega)$	$H^2(\mathcal{T}_h)$	H^2	h	yes

Table 5.1: Error rates for conforming P_1 elements, with $0 < r, s \leq 1$.

Dual regularity H^{1+s} means that every $g \in L^2(\Omega)$ gives a dual solution $z \in H^{1+s}(\Omega) \cap V$ with

$$\|z\|_{H^{1+s}(\Omega)} \leq C_{\mathrm{reg}} \|g\|_{L^2(\Omega)}.$$

The case $s = 1$ is the H^2 regularity assumption in Lemma 5.16. For a general solution $u \in V$, Lemma 5.14 still gives convergence in H^1 as $h \to 0$, without a prescribed algebraic rate.